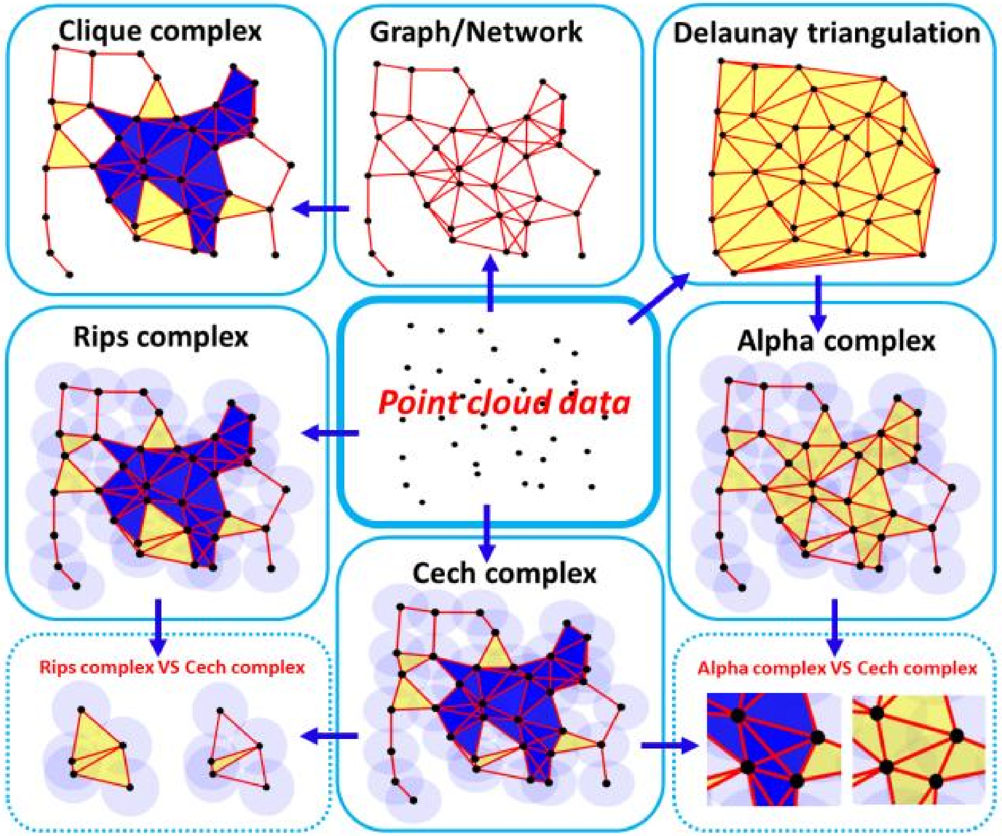
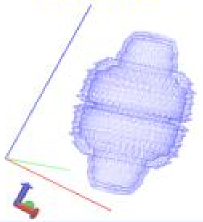
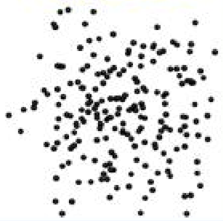
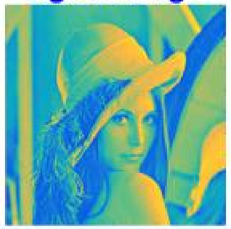
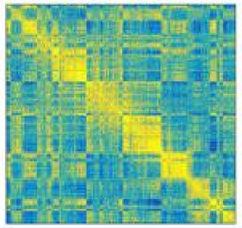
Algebra, Geometry and Topology-based Data Analysis and Deep Learning

Yu Chen

School of Mathematics Science, Zhejiang University

Real-world Data Representation



Function	Point Cloud	Digital image	Distance matrix	Network
				
Morse, Cubical	Rips, Alpha	Morse, Cubical	Rips, Clique	Rips, Clique
PERSEUS (Morse), PHAT (Morse), Dipha (Morse), GUDHI (Morse), R-TAD (Morse)	JAVAPLEX (Rips), PERSEUS (Rips), PHAT(Rips), Dipha (Rips), GUDHI (Rips), R-TAD (Rips, Alpha), DIONYSUS (Rips, Alpha)	PERSEUS (Morse), PHAT (Morse), Dipha (Morse), GUDHI (Morse), R-TAD (Morse)	JAVAPLEX (Rips), PERSEUS (Rips), PHAT (Rips), Dipha (Rips), GUDHI (Rips), R-TAD (Rips, Alpha), DIONYSUS (Rips), jHoles (Clique)	JAVAPLEX (Rips), PERSEUS (Rips), PHAT (Rips), Dipha (Rips), GUDHI (Rips), R-TAD (Rips, Alpha), DIONYSUS (Rips), jHoles (Clique)

浙江大學
ZHEJIANG UNIVERSITY

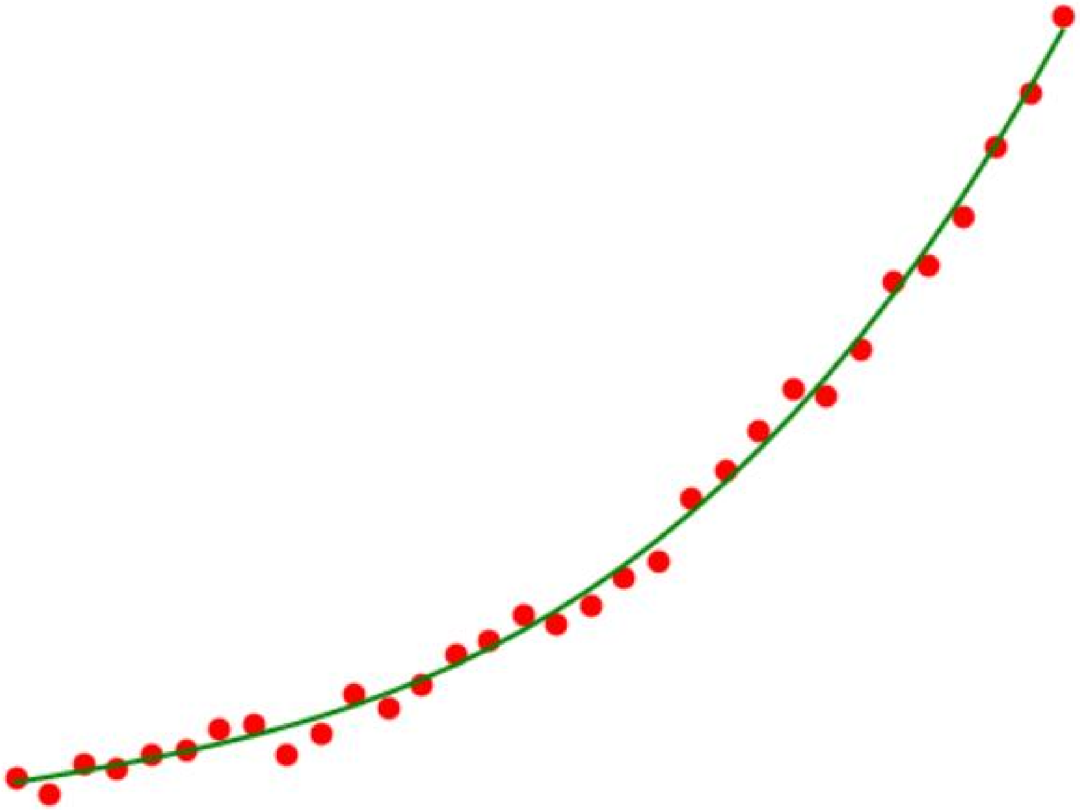
目录
CONTENTS

01

Persistent Homology :

how to capture “hole” structure?

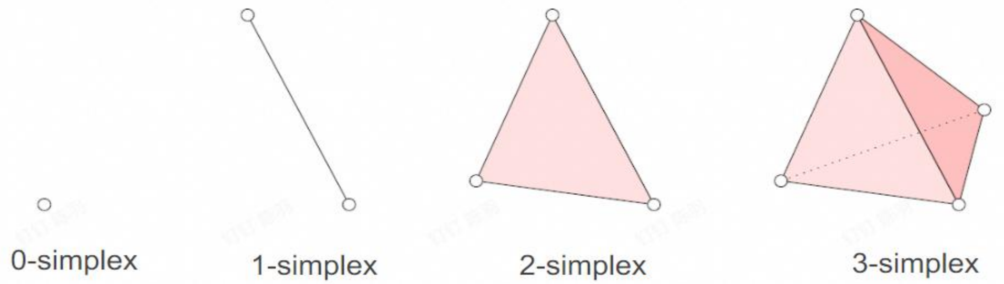
New Perspective in Data Analysis

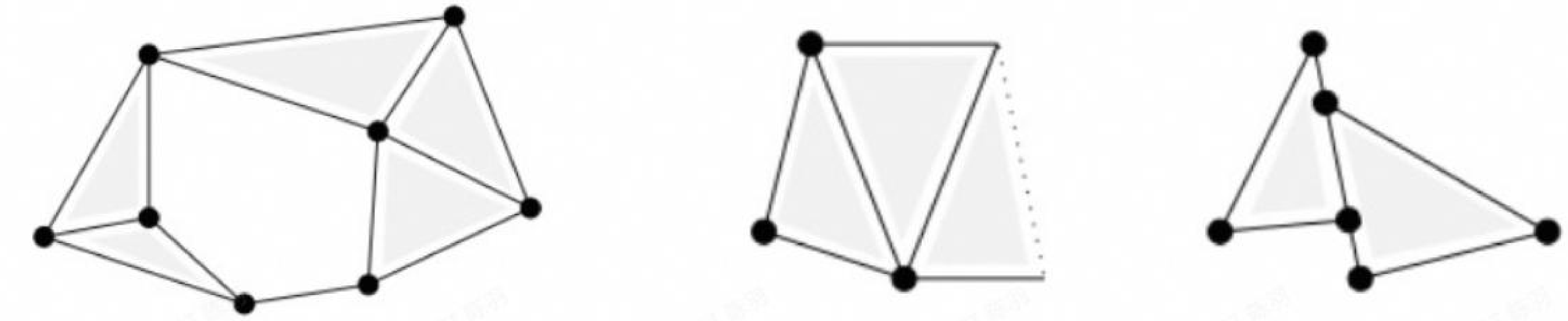


Simplicial Complex

- The convex hull of affine independent $k+1$ points is called a k-simplex.
- A simplicial complex K is a finite set that composed of simplices satisfying:

1. $\sigma \in K \; and \; \tau < \sigma \;\Rightarrow \tau \in K.$
2. $\sigma_1, \sigma_2 \in K \; and \; \sigma_1 \cap \sigma_2 \neq \emptyset \;\Rightarrow\; \sigma_1 \cap \sigma_2 < \sigma_1 \; and \; \sigma_1 \cap \sigma_2 < \sigma_2.$

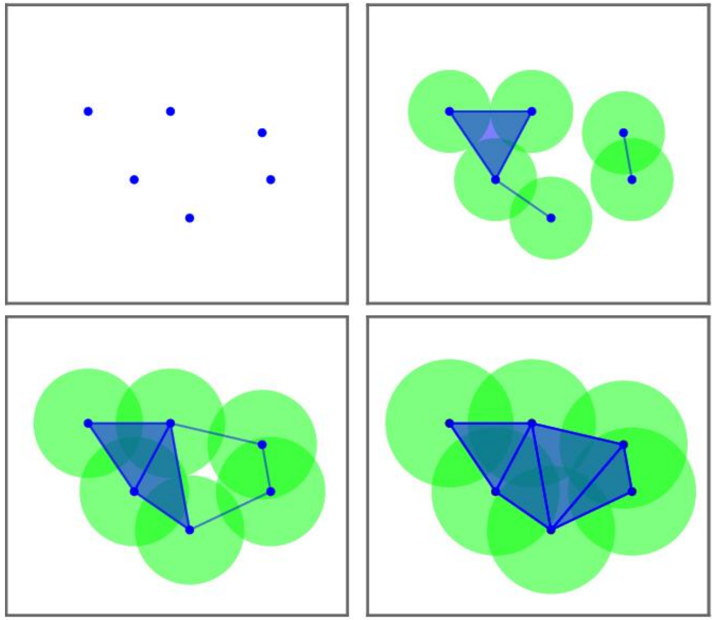


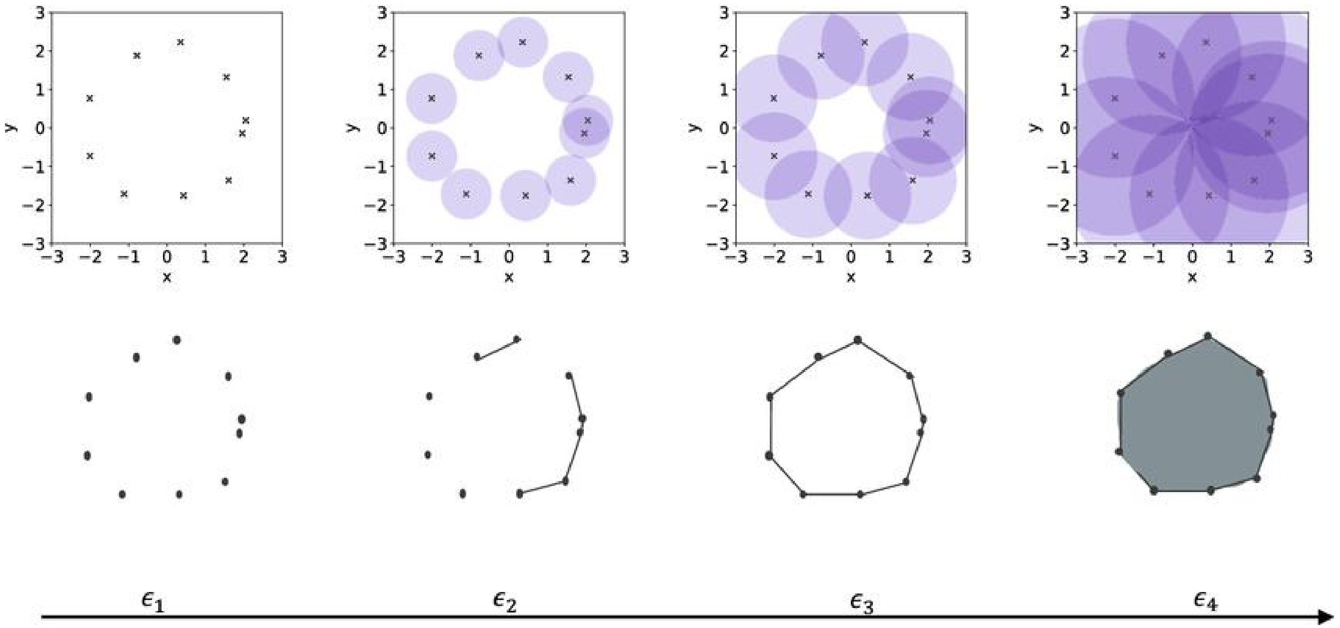


VR Complex and Filtration

- Suppose X is a finite point cloud in R^N, **Vietoris–Rips complex (VR complex)** is a kind of simplicial complex. Its construction is defined by the following rules: for any $\varepsilon > 0$, a finite subset $\{x_0, x_1, \ldots, x_n\} \subseteq X$ of R^N forms a simplex $[x_0, x_1, \ldots, x_n]$ if and only if the distance between any pair of points x_i and x_j satisfies $d(x_i, x_j) \leq \varepsilon$.
- Therefore a VR complex is constructed based on the distance between points.
- Given parameters $0 = \varepsilon_0 < \varepsilon_1 < \cdots < \varepsilon_m$, we can construct a VR filtration from X:

$$VR(X, \varepsilon_0) \subset VR(X, \varepsilon_1) \subset \cdots \subset VR(X, \varepsilon_m)$$





求是創新

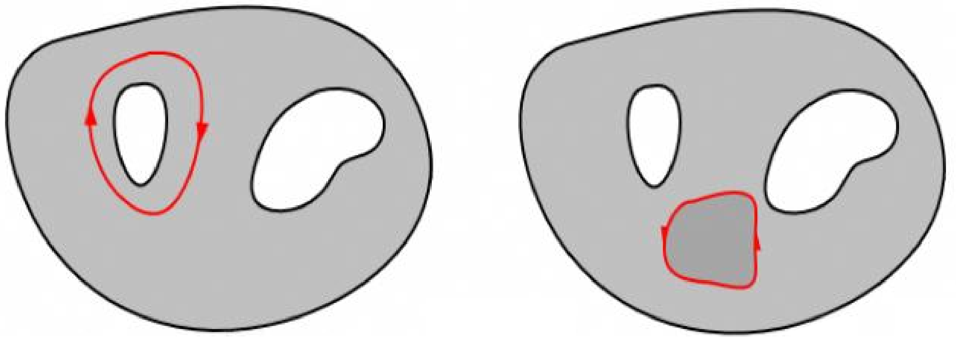
Homology Groups

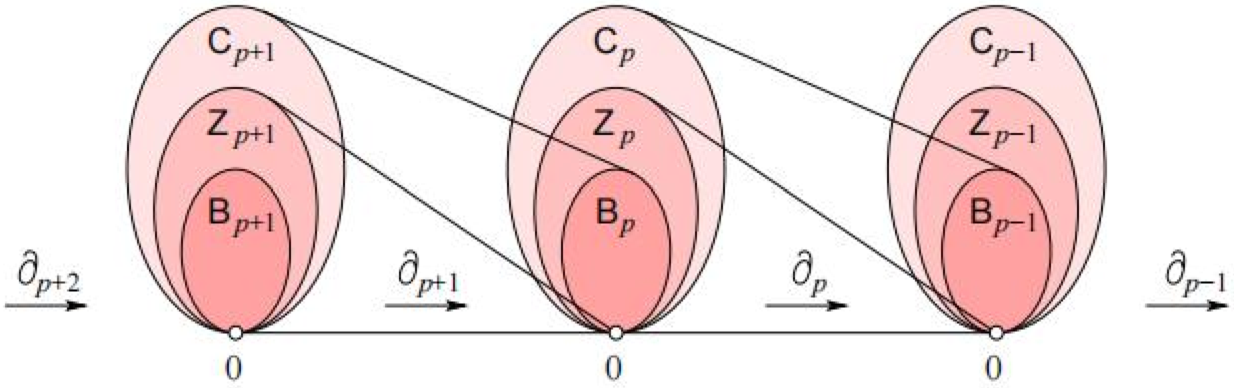
- Homology groups can be used to describe "holes" in space.
- The rank of the 0-dimensional homology group describes the number of connected components (0-dimensional holes). The rank of the 1-dimensional homology group describes the number of 1-dimensional loops (1-dimensional holes). The rank of the 2-dimensional homology group describes the number of 2-dimensional voids (2-dimensional holes), etc.

$$[v_0, v_1 \dots v_k] \mapsto \sum_{i=0}^{k} (-1)^i [v_0, \dots \widehat{v_i} \dots, v_k]$$

$$Z_k(\mathcal{K}) = \ker \partial_k \qquad B_k(\mathcal{K}) = \operatorname{im} \partial_{k+1}$$

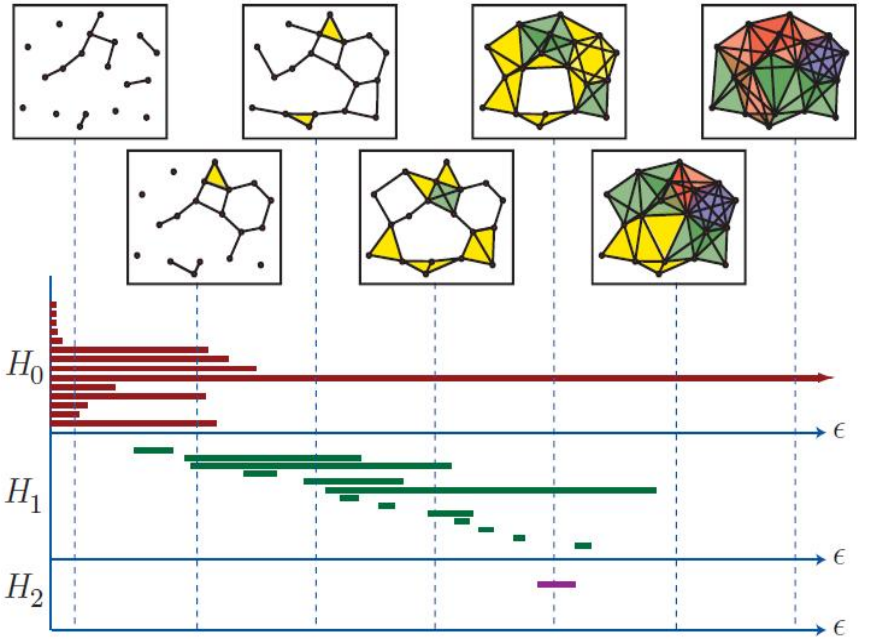
$$H_k(\mathcal{K}) = Z_k(\mathcal{K}) / B_k(\mathcal{K})$$

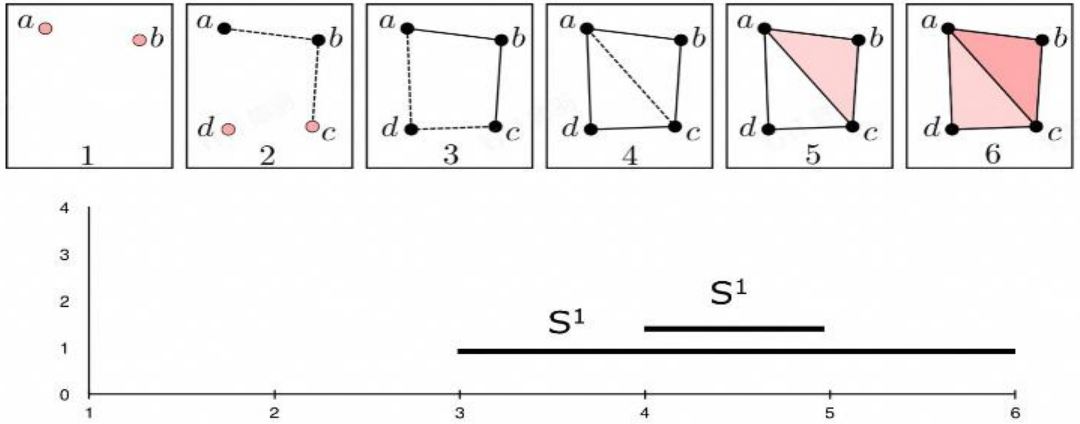




Persistent Homology

- **Homology groups**
  - Homology groups can be used to describe "holes" in space.
  - The rank of the k-dimensional homology group is called the k-th Betti number.
- **Persistent Homology Groups**
  - Persistent homology study the changes in homology groups within a filtration.
  - Describe the intrinsic topological structure of the underlying space (e.g., a point cloud) that generates the filtration.

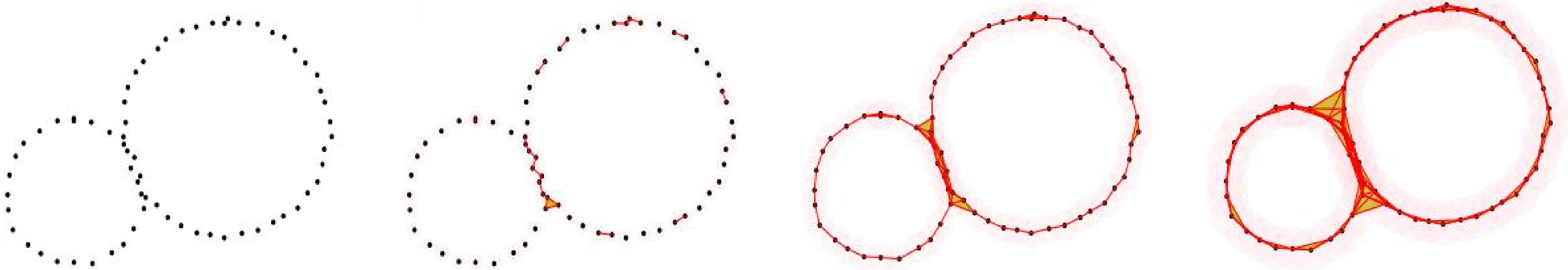


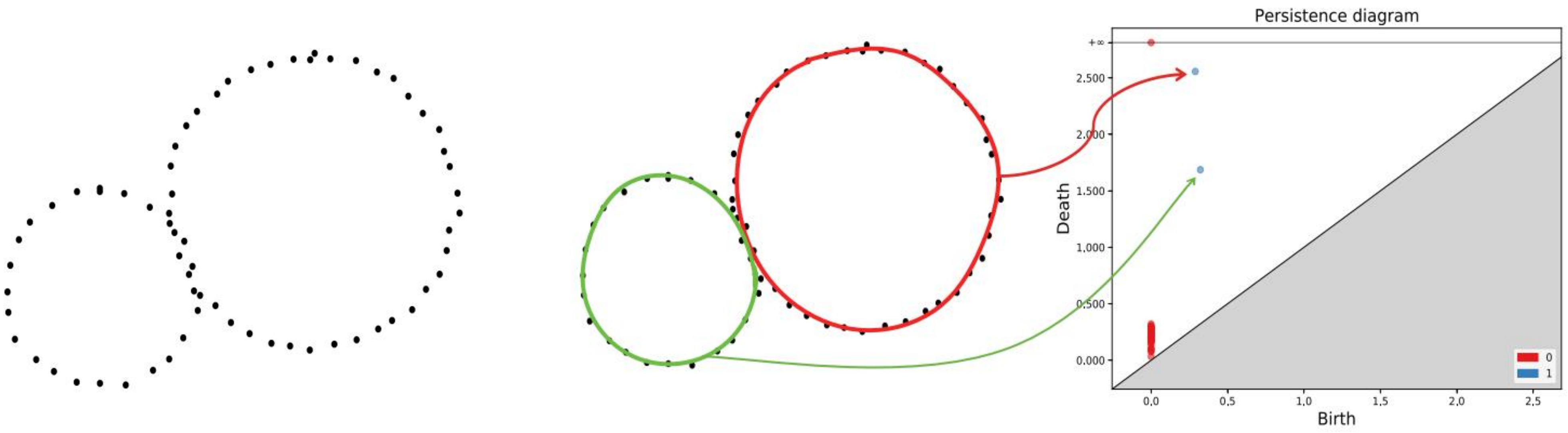


求是創新

Persistent Homology

➢ **Persistence diagram (PD)**

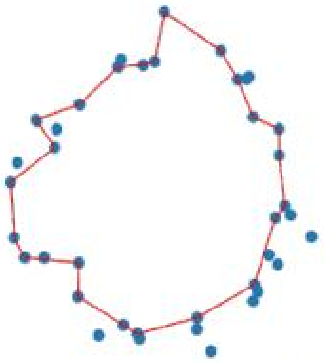




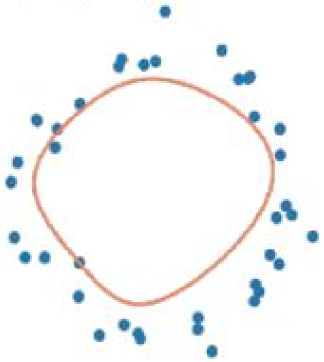
Applications: Topology-aware Geometric Reconstruction

Algorithm outline:

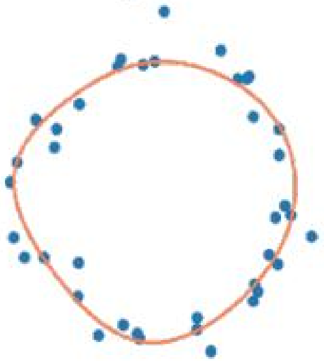
(a) Input point cloud.



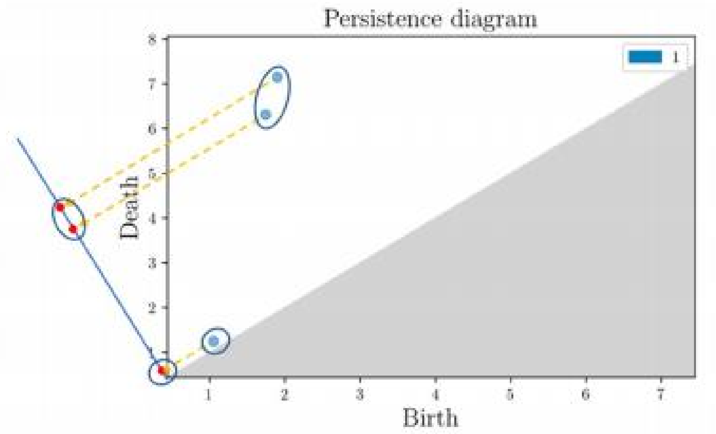
(b) Extracted persistent 1-cycle.

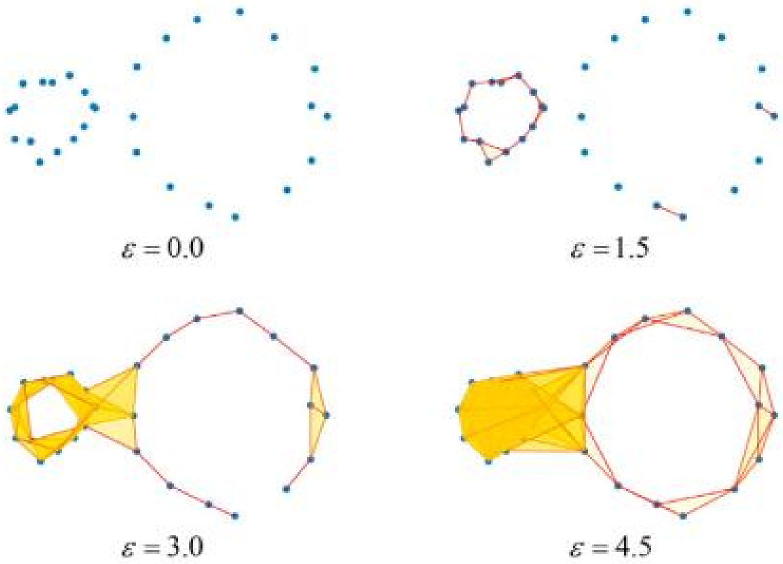


(c) Initial parametric B-spline curve



(d) Reconstructed parametric B-spline curve.



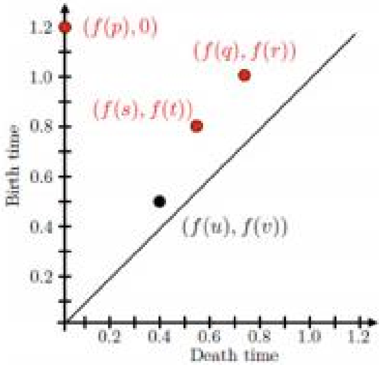


Applications: Topology-aware Geometric Reconstruction

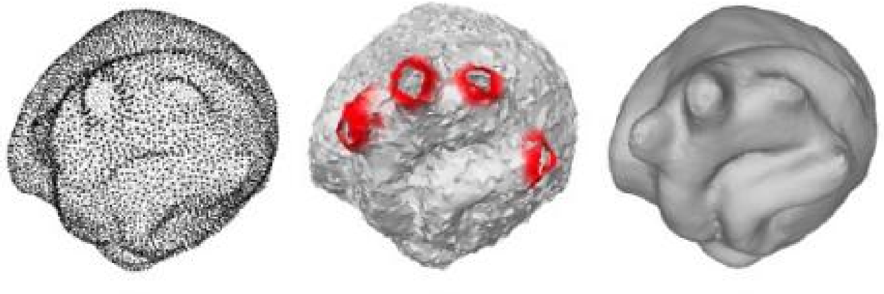
- **Objective Function:**

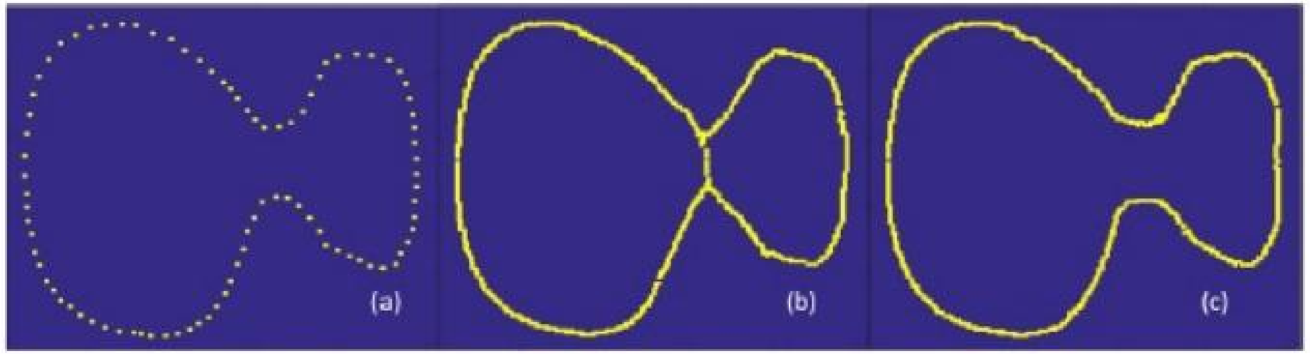
$$-((d_1-b_1)^2-(d_2-b_2)^2)$$

$$-((d_2-b_2)^2-(d_3-b_3)^2)$$



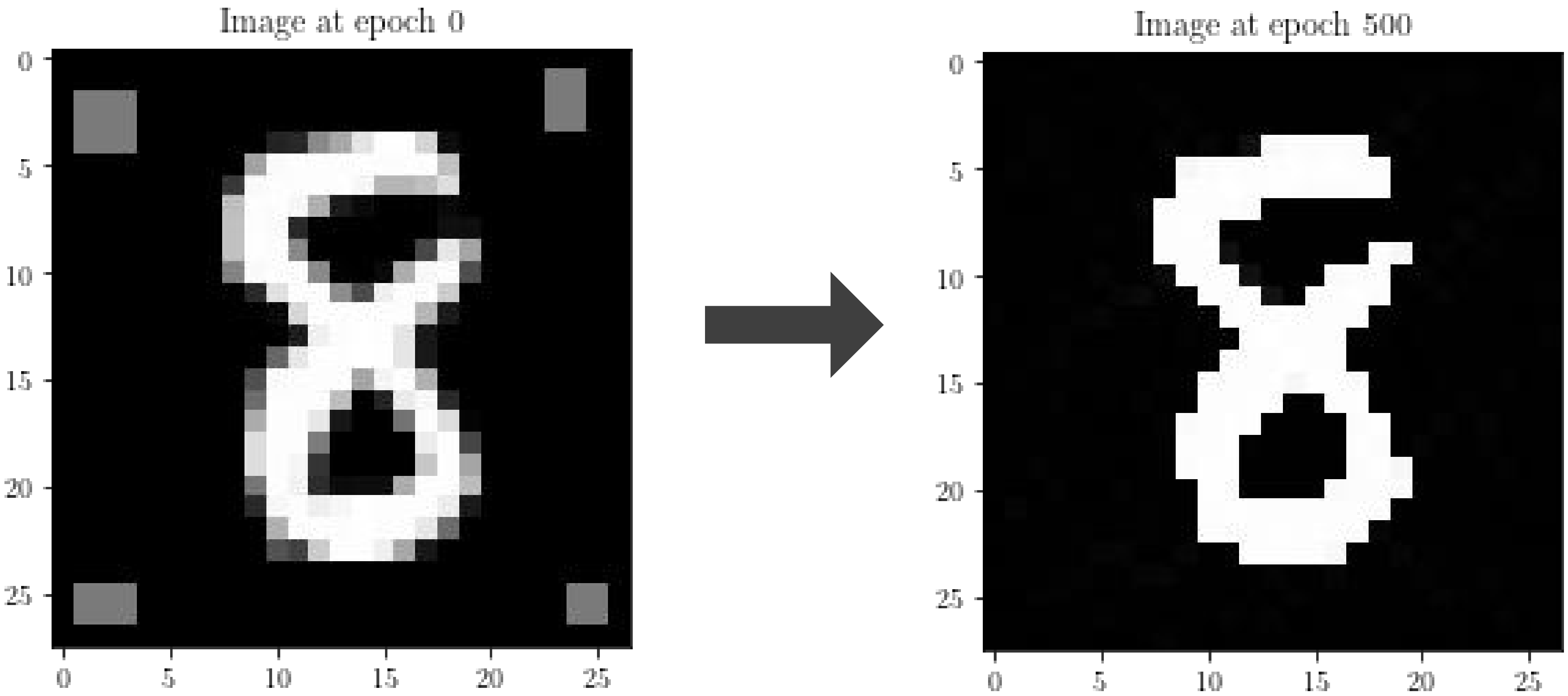

(a) (b) (c)


(a) (b) (c)

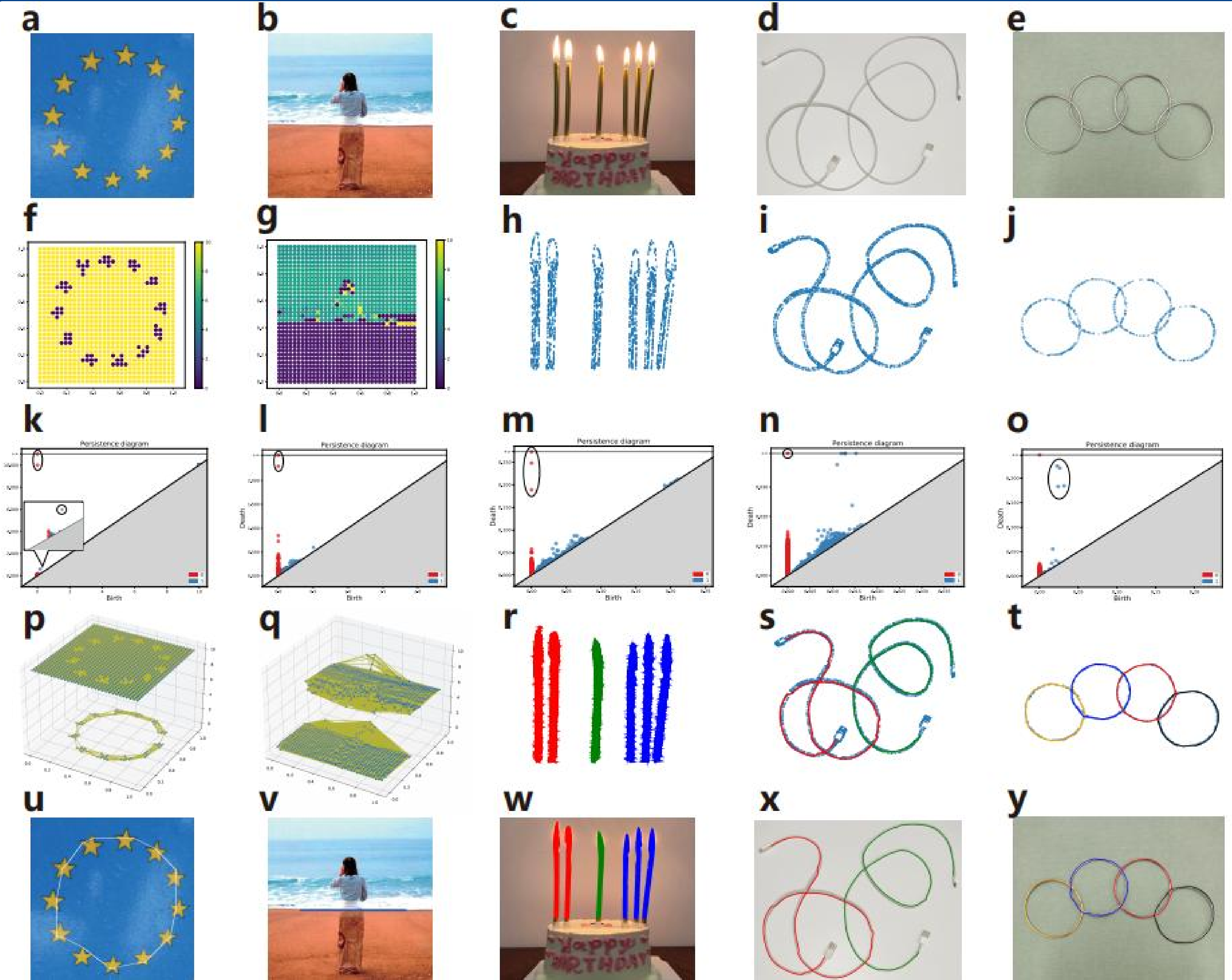


Applications: Image Processing

➢ The left image contains four stains. Since those stains correspond to connected components, we can detect them with 0-dimensional persistent homology, and try to remove them by optimizing the pixel values so that the points in the 0-dimensional persistence diagram have minimal distances to the diagonal.

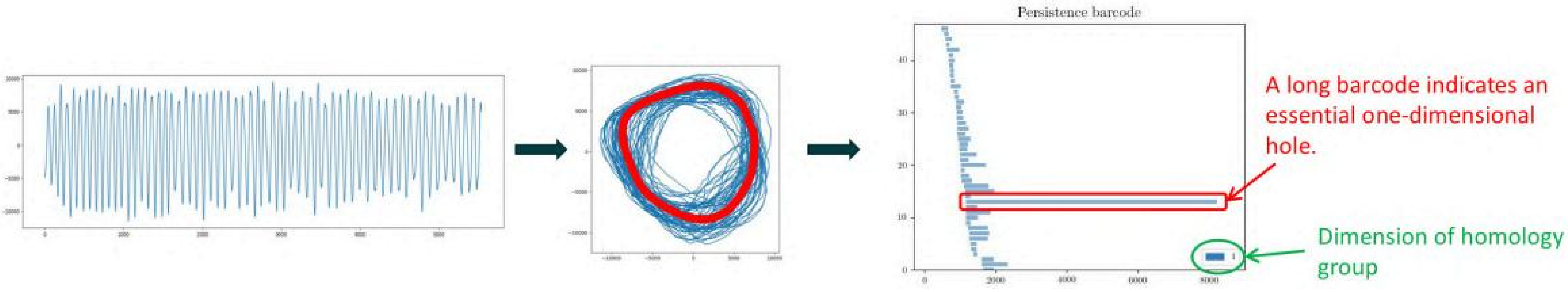


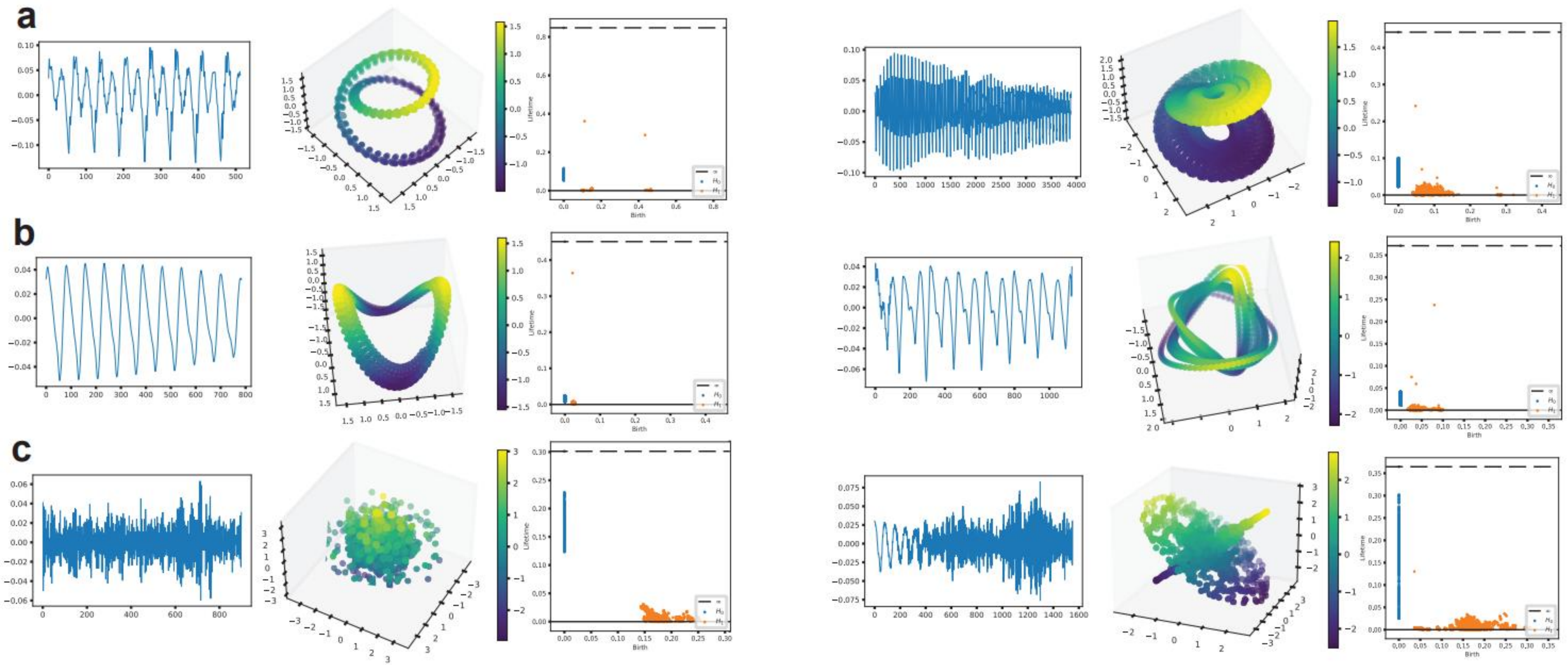
Applications: Image Processing



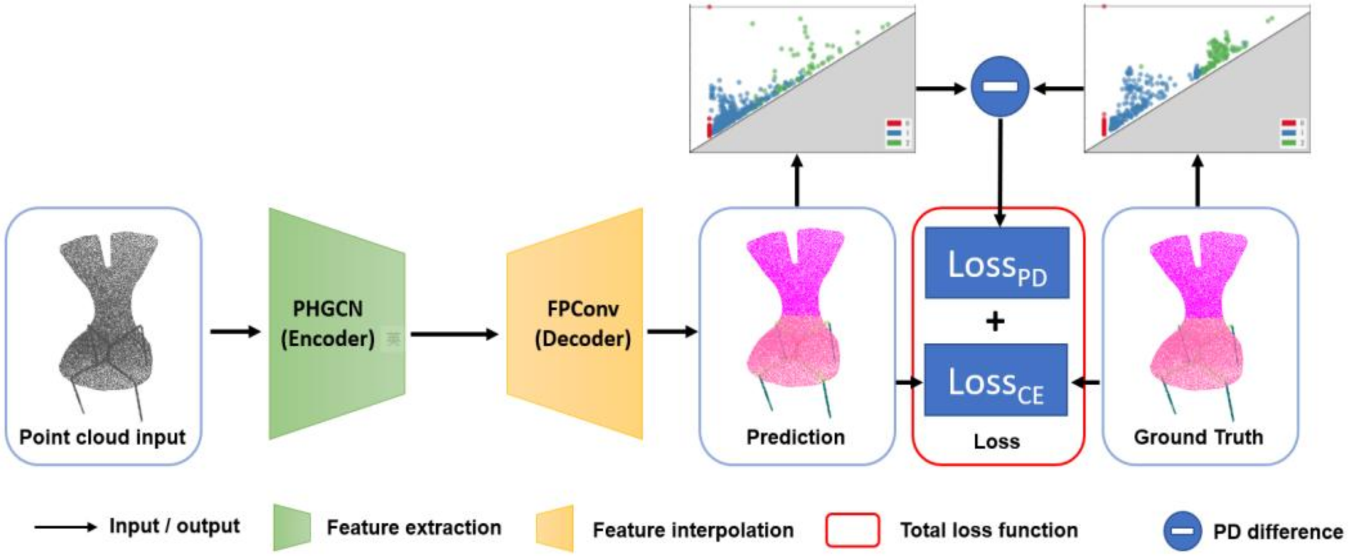
求是创新

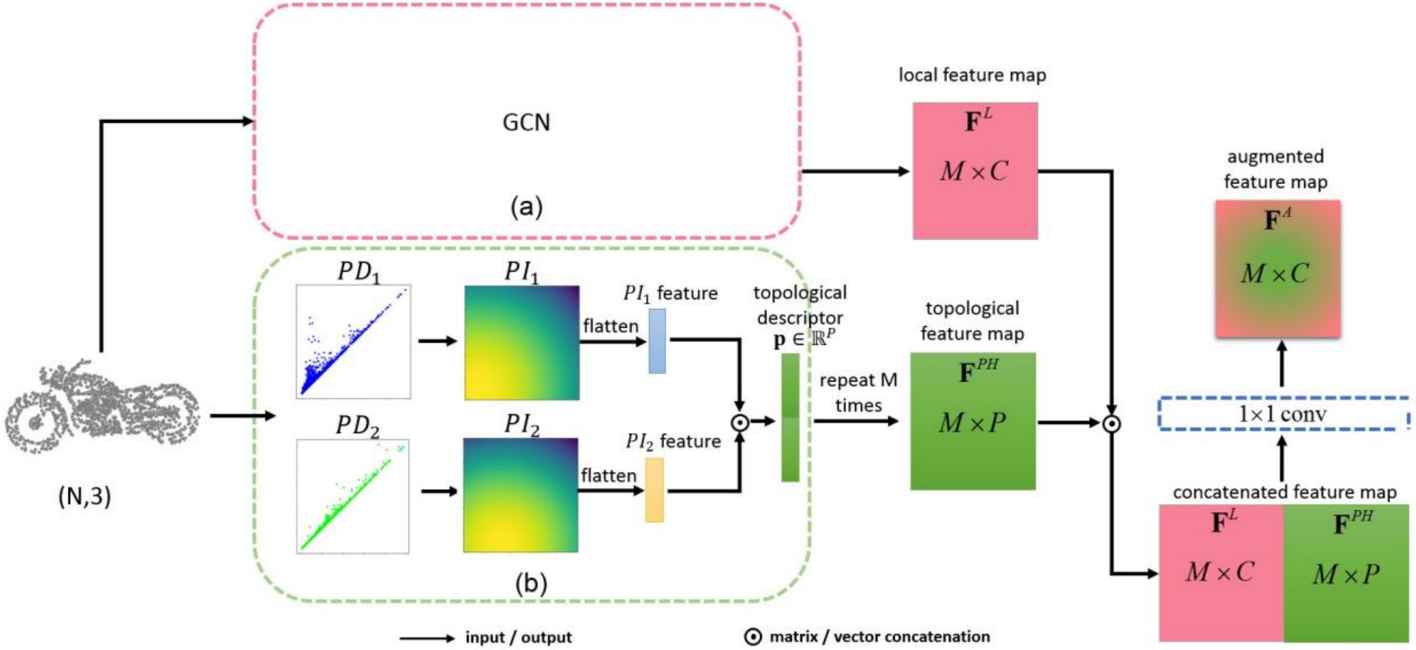
Applications: Time Sequence Analysis





PHGCN (ICCV 2021)





浙江大學 ZHEJIANG UNIVERSITY

Homologies of path complexes and digraphs

Alexander Grigor'yan *
Department of Mathematics
University of Bielefeld
33613 Bielefeld, Germany

Yong Lin †
Department of Mathematics
Information School
Renmin University of China
Beijing 100872, China

Yuri Muranov‡
Department of Mathematics
Grodno State University
Grodno 230023, Republic of Belarus

Shing-Tung Yau
Department of Mathematics
Harvard University
Cambridge MA 02138, USA

May 2013

Contents

02

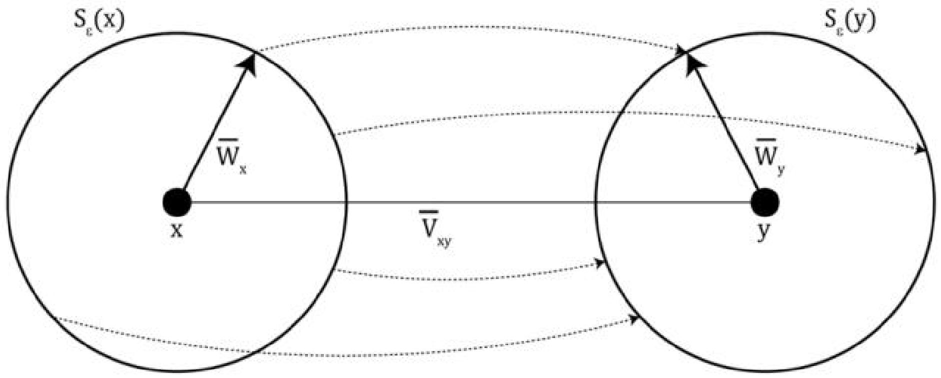
Ricci Curvature and Ricci Flow: the geometry of graphs

Ricci Curvature

- Ricci curvature can measure the growth of volumes of distance balls, transportation distances between balls, divergence of geodesics. On two dimensional manifold, Ricci curvature equals to the classical Gauss curvature.
- Given two close points x and y in a Riemannian manifold of dimension n, defining a tangent vector v_{xy}, one can consider the parallel transport in the direction v_{xy}. Then points on an infinitesimal sphere $S_\varepsilon(x)$ centered at x, are transported to points y on the corresponding sphere $S_\varepsilon(y)$ by an average distance equal to

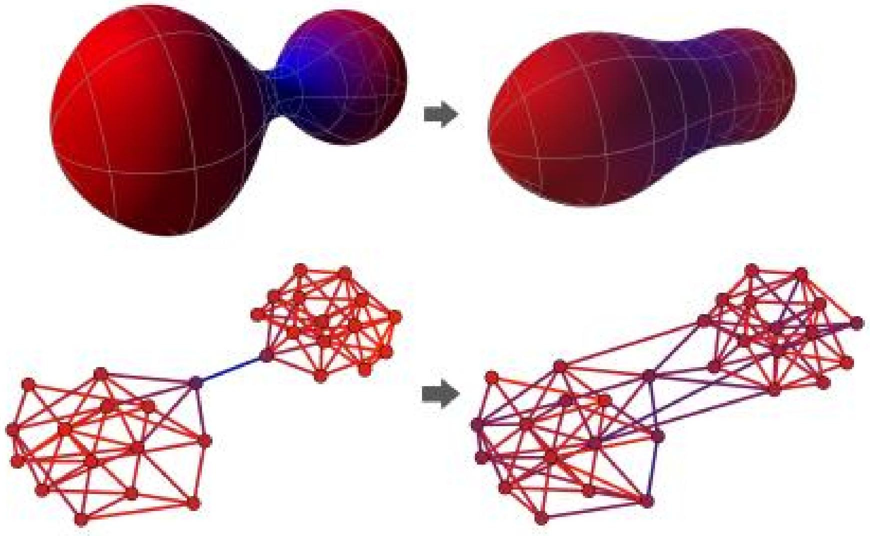
$$d(x,y)(1-\frac{\varepsilon^2}{2n}Ric(v_{xy}))$$

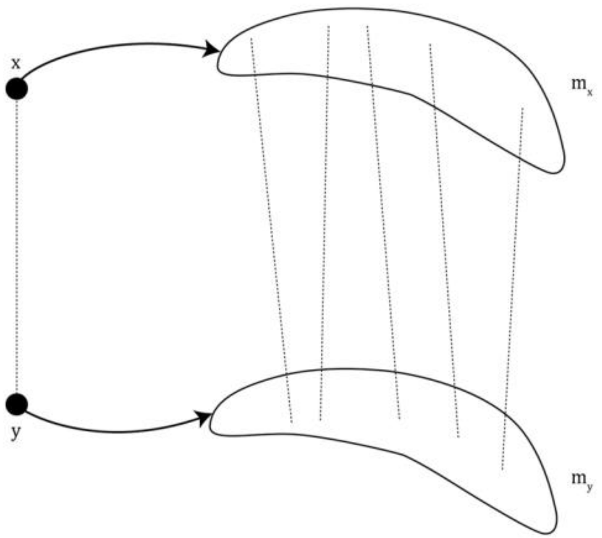
- In Riemannian manifolds of positive (respectively, negative) curvature, balls are closer (respectively, farther) than their centers. Thus, in spaces of positive Ricci curvature spheres are closer than their centers, while in spaces of negative curvature they are farther away.



Discrete Ricci Curvature

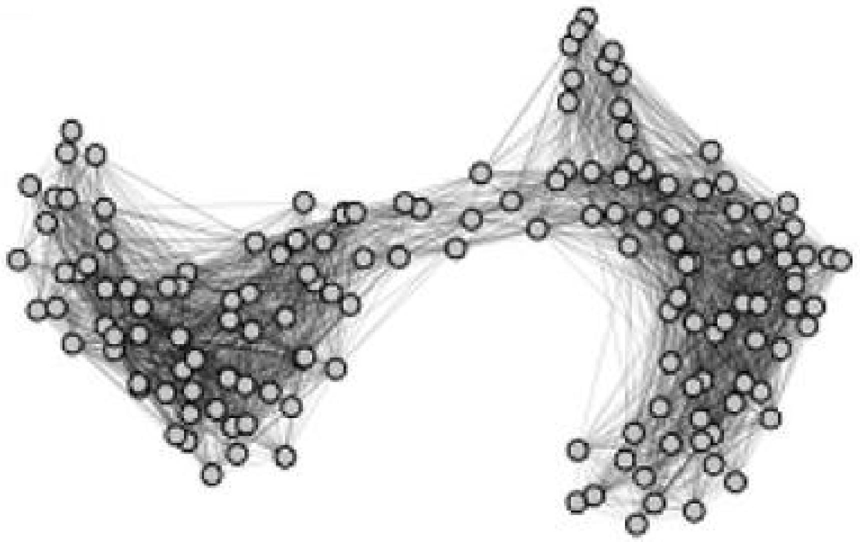
- To generalize idea on Riemannian manifold to metric measure spaces, one has to replace the (volumes of) spheres or balls, by measures m_x, m_y. Points will be transported by a distance equal to $(1-k)d(x,y)$, on the average, where $k = k(x,y)$ represents the coarse (Ollivier) curvature along the geodesic segment xy.
- On graph or network, discrete Ricci curvatures have been used in the characterization of "over-squashing" phenomenon, which happens at the bottleneck region of a network when the messages propagated from distant nodes distort significantly.
- In order to use Ricci curvature to analyze real-world data, discrete Ricci curvature forms, i.e., Ollivier Ricci curvature (ORC) and Forman Ricci curvature (FRC), have been developed.

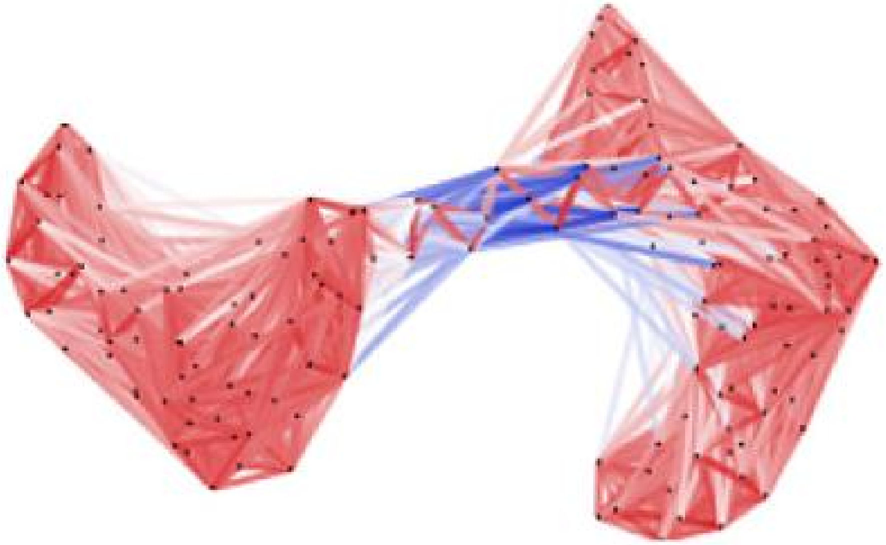




Ollivier Ricci Curvature

- Ollivier-Ricci curvature is a discrete Ricci curvature model that is developed for the analysis of graph data. ORC has been combined with deep learning models and demonstrated great advantages.
- ORC is defined on graph edges. It measures the difference between the edge "distance" (or length of edge) and transportation distance of two probability distributions.
- Positive edge ORC means that there are strong connections (or short "distance") between the two respective neighborhoods, and negative edge ORC indicates weak connections (or long "distance").





Ollivier Ricci Curvature

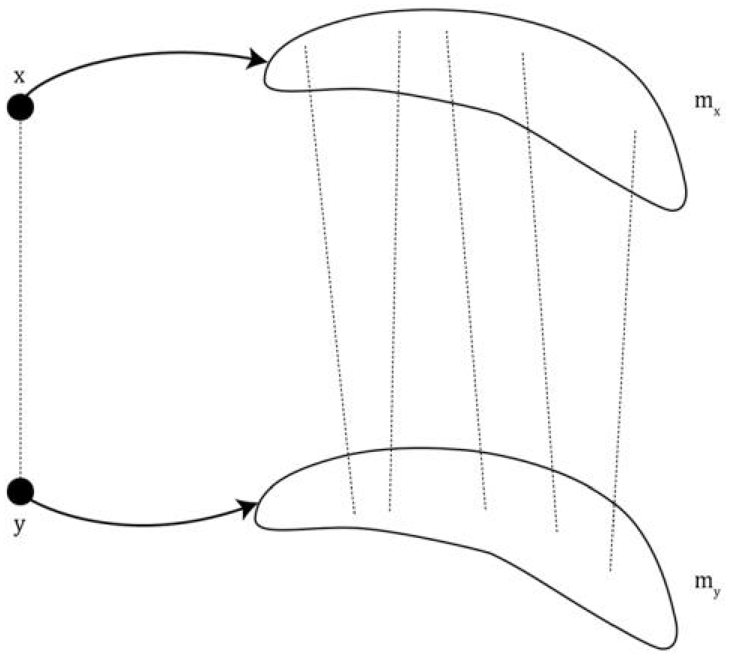
- Ollivier-Ricci curvature measures the difference between the edge “distance” (or length of edge) and transportation distance of two probability distributions：

$$\kappa(x, y) = 1 - \frac{W_1(m_x, m_y)}{d(x, y)}$$

$$W_1(m_x, m_y) = \inf_{\mu_{x,y} \in \prod(m_x, m_y)} \sum_{(x', y') \in V \times V} d(x', y') \mu_{x,y}(x', y')$$

$$\sum_{y' \in V} \mu_{x,y}(x', y') = m_x(x'), \quad \sum_{x' \in V} \mu_{x,y}(x', y') = m_y(y')$$

$$m_{v_i}(v_j) = \begin{cases} \alpha & \text{if } v_j = v_i \\ (1-\alpha)/k_{v_i} & \text{if } v_j \in \mathcal{N}(v_i) \\ 0 & \text{otherwise} \end{cases},$$

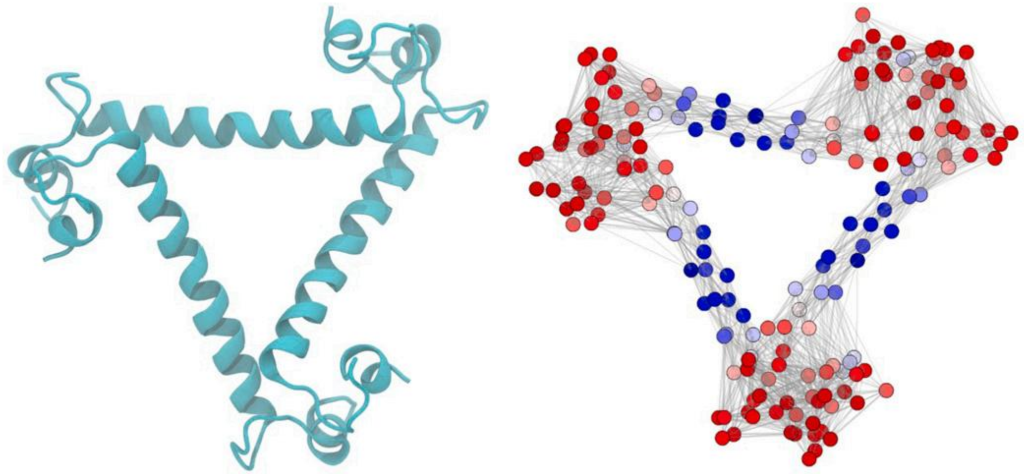


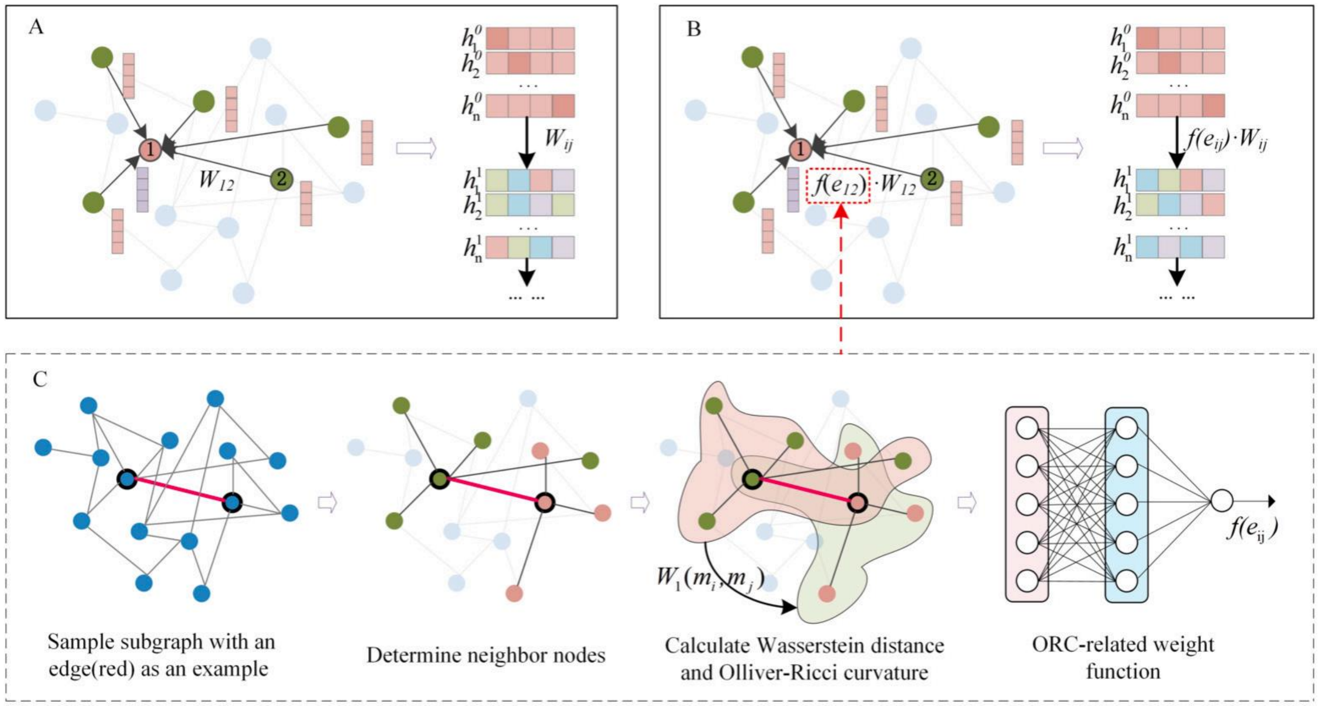
- ORCs are calculated on edges of graphs, and the node ORCs are usually defined as the average of edges ORCs.

Ricci Curvature-based Deep Learning

- The (Ollivier) Ricci curvature on the graph can be an important feature of the graph, which describes the "bottleneck" phenomenon in the graph.
- Assigning curvature to each edge and using curvature as the main feature for deep learning (MLP, GCN, etc.) can effectively handle many real-word problems such as protein structure prediction.

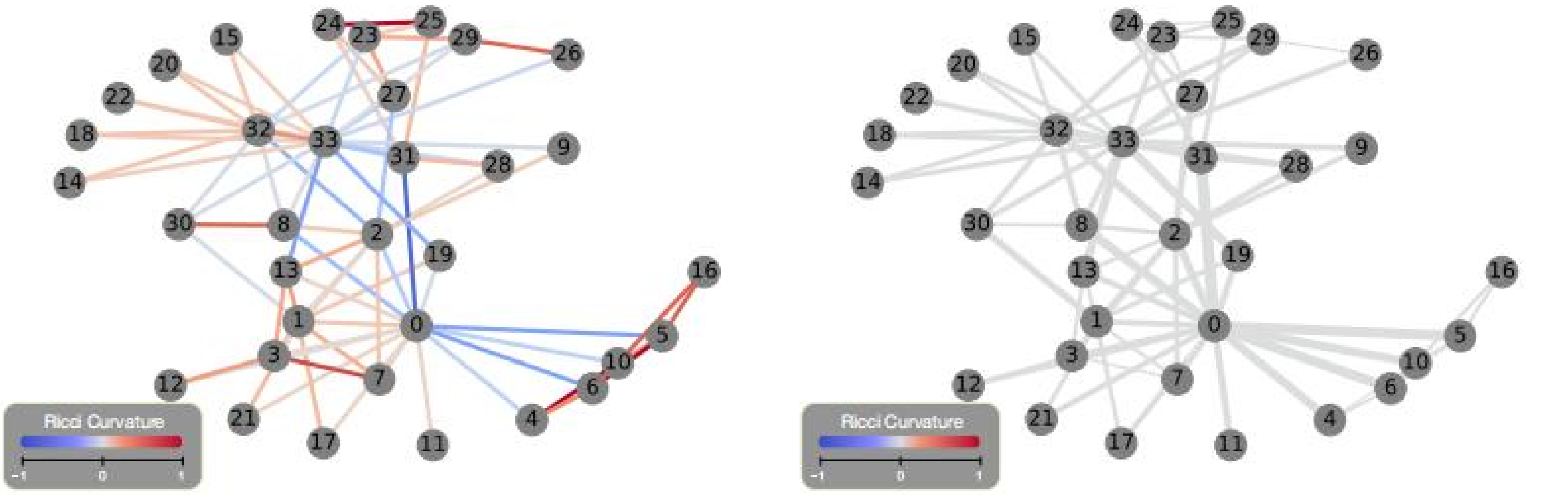
$$h_v^{(l+1)} = \sigma\left(\sum_{v_j \in \mathcal{N}(v) \cup \{v\}} \frac{1}{\sqrt{\deg(v)}\sqrt{\deg(v_j)}} f_c(v, v_j) W_{\mathrm{GCN}} h_{v_j}^{(l)} \right)$$





Ricci Flow

- Ricci flow is a process that deforms the metric while the Ricci curvature evolves to be uniform everywhere.
- A surface Ricci flow is the process to deform the Riemannian metric of the surface, proportional to the Gaussian curvature, such that the curvature evolves like heat diffusion and becomes uniform at the limit. Intuitively, this behaves as flattening a piece of crumpled paper.
- Graph Ricci flow deformed the edge weights until the edge Ricci curvatures converged.



(a) Ricci Curvature: The Original (b) Ricci Curvature: After Ricci Flow

Ricci Flow

- Graph Ricci flow deformed the edge weights until the edge Ricci curvatures converged.
- For any pair of adjacent nodes x and y on a graph $G = (V, E)$, we adjust the edge weight of $\overline{xy}$, $w(x, y)$, by the (Ollivier) Ricci curvature $\kappa(x, y)$. Ricci flow deformed the edge weights until Ricci curvatures converged.

$$w_{i+1}(x, y) = w_i(x, y) - \epsilon \cdot \kappa_i(x, y) \cdot w_i(x, y), \quad \forall \overline{xy} \in E$$

Algorithm 1: Ricci flow on a graph

Input : An undirected graph G.

Output: A weighted graph G in which the Ricci curvature on all edges are the same.

1 Initialize $w_0(x, y) = 1$, $\forall \overline{xy} \in E$;

2 Compute the Ricci curvature $\kappa_0(x, y)$, $\forall \overline{xy} \in E$;

3 Update the edge weight by

$$w_{i+1}(x, y) \leftarrow w_i(x, y) - \epsilon \cdot \kappa_i(x, y) \cdot w_i(x, y);$$

4 Normalize the edge weight by

$$w_{i+1}(x, y) \leftarrow w_{i+1}(x, y) \cdot \frac{|E|}{\sum_{\overline{xy} \in E} w_{i+1}(x, y)};$$

5 Repeat $2 - 4$ until the curvature values do not change much.

RicciNets (ICML 2020)

RicciNets: Curvature-guided Pruning of High-performance Neural Networks Using Ricci Flow

S. Glass SEG67@CAM.AC.UK
Department of Physics, University of Cambridge

S. Spasov SES88@CAM.AC.UK
Department of Computer Science and Technology, University of Cambridge

P. Liò PL219@CAM.AC.UK
Department of Computer Science and Technology, University of Cambridge

Abstract

A novel method to identify salient computational paths within randomly wired neural networks *before* training is proposed. The computational graph is pruned based on a node mass probability function defined by local graph measures and weighted by hyperparameters produced by a reinforcement learning-based controller neural network. We use the definition of Ricci curvature to remove edges of low importance before mapping the computational graph to a neural network. We show a reduction of almost 35% in the number of floating-point operations (FLOPs) per pass, with no degradation in performance. Further, our method can successfully regularize randomly wired neural networks based on purely structural properties, and also find that the favourable characteristics identified in one network generalise to other networks. The method produces networks with better performance under similar compression to those pruned by lowest-magnitude weights. To our best knowledge, this is the first work on pruning randomly wired neural networks, as well as the first to utilize the topological measure of Ricci curvature in the pruning mechanism.

1. Introduction

At birth, the construction of the most important networks is largely random and random graph modelling is heavily used in the study of the human brain (Bullmore and Sporns, 2009; Bassett and Sporns, 2017). Recent work on randomly wired neural networks has emulated this in the field of deep learning, and moves away from the wiring approach that has typically dominated NAS (Xie et al., 2019). Randomly wired networks display comparable performance to state-of-the-art architectures (eg. ResNet, DenseNet), and provide a relatively unrestricted space on which to perform further optimisation. We propose a search method that takes place within a low-dimensional search space; a pruning method-

Graph Geometric Representation (NeurIPS 2024)

浙江大學 ZHEJIANG UNIVERSITY

Exploring Geometric Representational Alignment through Ollivier-Ricci Curvature and Ricci Flow

Nahid Torbati TORBATI@CBS.MPG.DE
Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany

Michael Gaebler GAEBLER@CBS.MPG.DE
Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany
Berlin School of Mind and Brain, Humboldt-Universität zu Berlin, Berlin, Germany

Simon M. Hofmann* SIMON.HOFMANN@CBS.MPG.DE
Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany

Nico Scherf* NSCHERF@CBS.MPG.DE
Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany
Center for Scalable Data Analytics and Artificial Intelligence (ScaDS.AI), Dresden/Leipzig, Germany
* *equal contribution.*

Abstract

Representational analysis explores how input data of a neural system are encoded in high-dimensional spaces of its distributed neural activations, and how we can compare different systems, for instance, artificial neural networks and brains, on those grounds. While existing methods offer important insights, they typically do not account for local intrinsic geometrical properties within the high-dimensional representation spaces. To go beyond these limitations, we explore Ollivier-Ricci curvature and Ricci flow as tools to study the alignment of representations between humans and artificial neural systems on a geometric level. As a proof-of-principle study, we compared the representations of face stimuli between VGG-Face, a human-aligned version of VGG-Face, and corresponding human similarity judgments from a large online study. Using this discrete geometric framework, we were able to identify local structural similarities and differences by examining the distributions of node and edge curvature and higher-level properties by detecting and comparing community structure in the representational graphs.

Keywords: Representational alignment, Ollivier-Ricci curvature, Ricci flow, Graph-based representation

求是創新

03

Laplacion Operator and Sheaf Theory: new geometry and topology descriptors of data

Laplacion Operator (Matrix)

- For a simplicial complex K, its combinatorial Laplacian matrix (Hodge Laplacian) $L_p = B_p^T B_p + B_{p+1} B_{p+1}^T$ where B_p is the p-th boundary matrix.

$$B_k(i,j) = \begin{cases} 1, & \text{if } \sigma_i^{k-1} \subset \sigma_j^k \text{ and } \sigma_i^{k-1} \sim \sigma_j^k \\ -1, & \text{if } \sigma_i^{k-1} \subset \sigma_j^k \text{ and } \sigma_i^{k-1} \nsim \sigma_j^k \\ 0, & \text{if } \sigma_i^{k-1} \not\subset \sigma_j^k \end{cases}$$

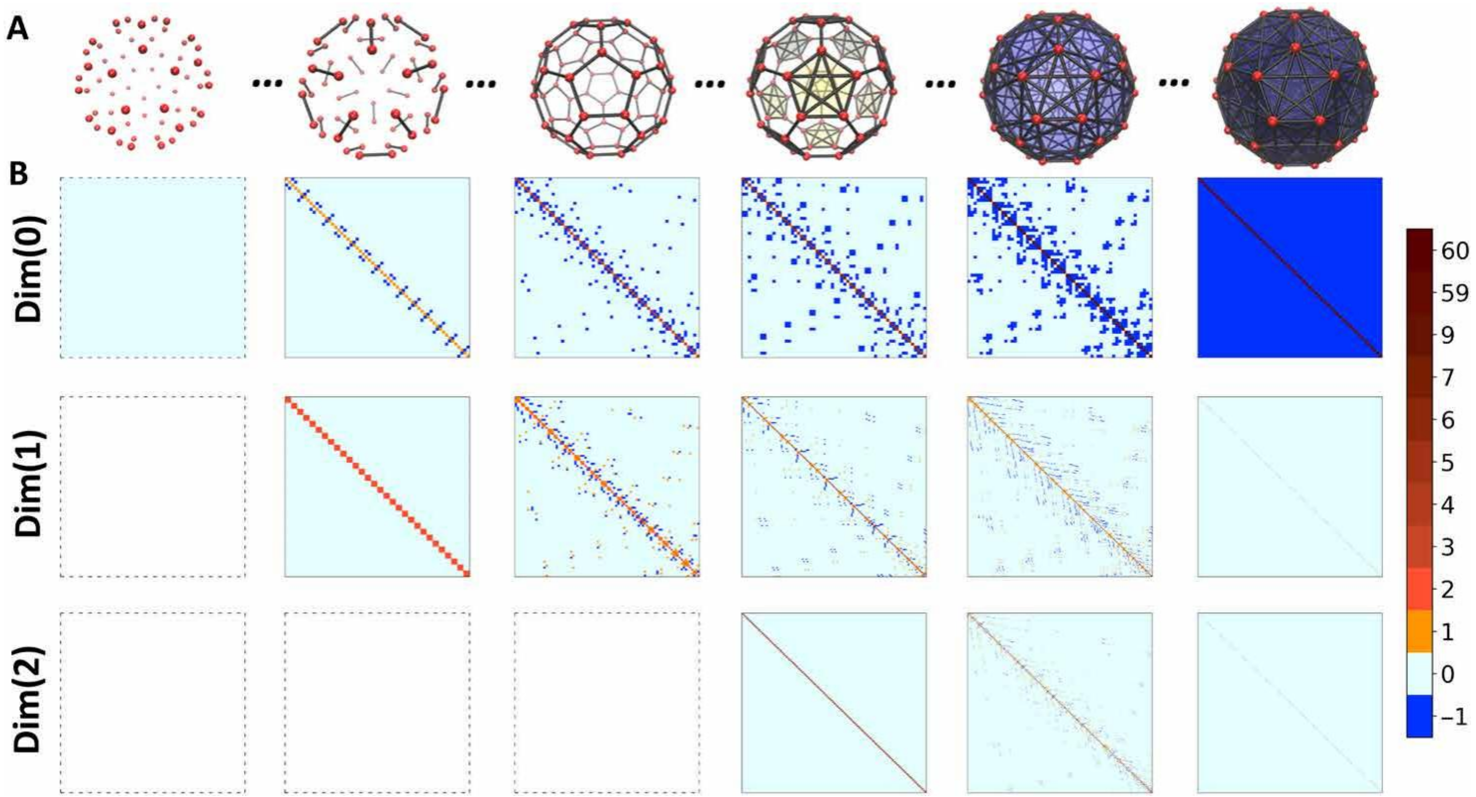
- If the simplicial complex is simply a graph, its Laplacian matrix can be written as

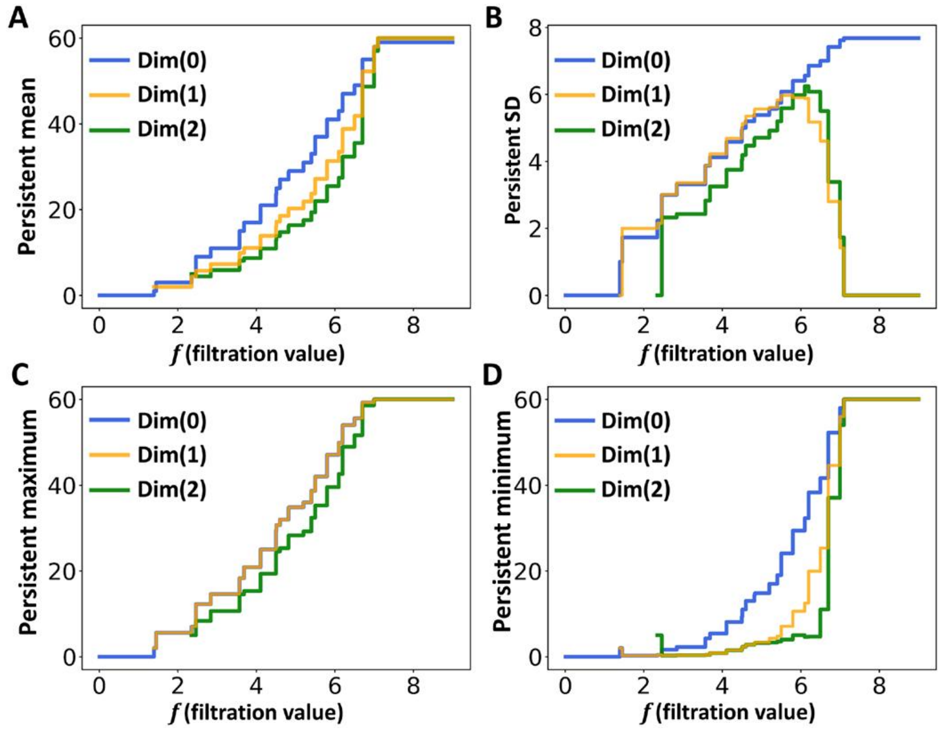
$$L(i,j) = \begin{cases} d(v_i), & i = j \\ -1, & i \neq j \text{ and} (v_i, v_j) \in E \\ 0, & i \neq j \text{ and} (v_i, v_j) \notin E \end{cases}$$

- Generally speaking, Laplacian matrices from graphs characterize relations between vertices, combinatorial Laplacian matrices from simplicial complexes describe connections between simplexes.

Laplacion Spectrum Analysis

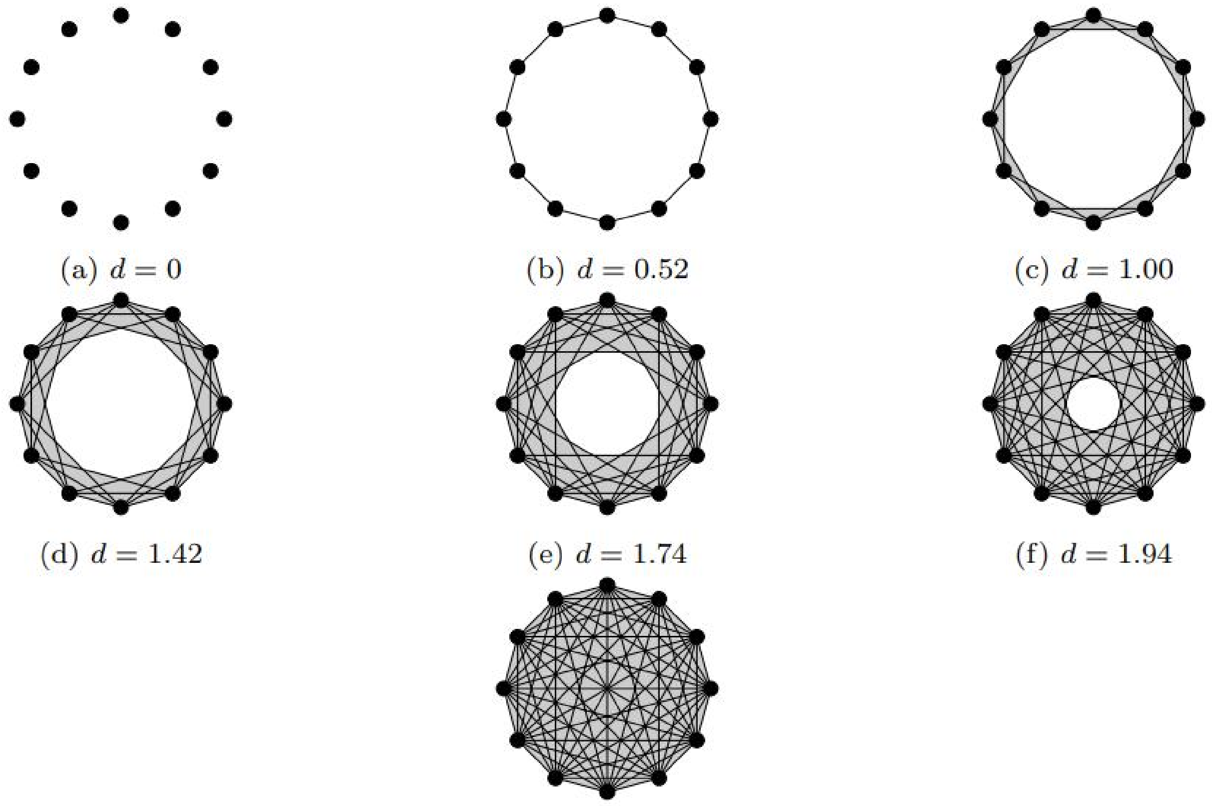
- The spectral information of Laplacion matrix includes eigenvalues, eigenvectors, characteristic polynomials, and eigenfunctions, etc.
- PerSpect (Persistent Laplacion Spectrum) theory studies the change of eigenspectrum when the structure representation, i.e., graph, simplicial complex, etc, grows from a set of isolated vertices to a fully connected topology, according to their inner structure connectivity and a predefined filtration parameter.





Laplacion Spectrum Analysis

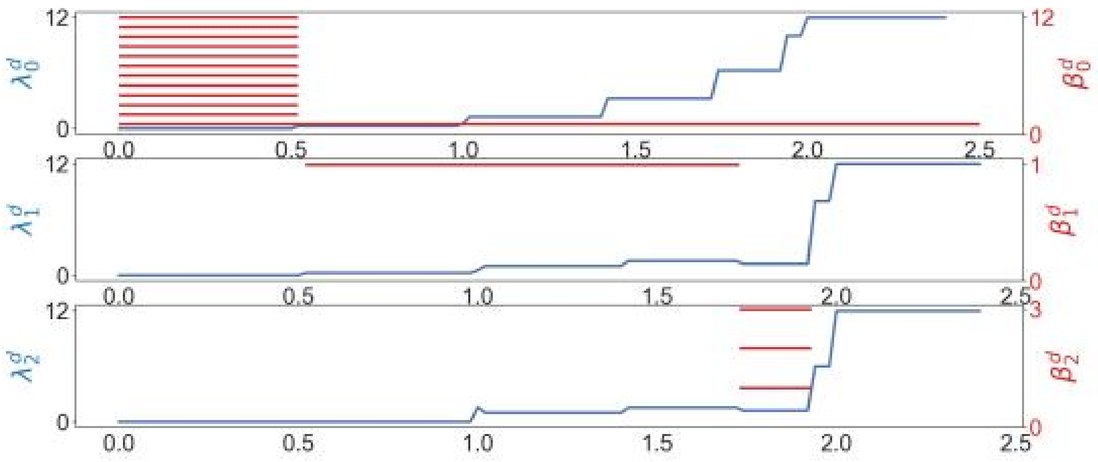
- Number of zero eigenvalues of q-th combinatorial Laplacian equals to the q-th Betti number (topology information), while non-zero eigenvalues shows information of geometry.
- Eigenvectors of zero eigenvalues have support near q-dimensional “holes” (or vertices in a connected component when q = 0); eigenvectors of nonzero eigenvalues have support near “clusters” of q-simplices.



(a) $d = 0$ (b) $d = 0.52$ (c) $d = 1.00$

(d) $d = 1.42$ (e) $d = 1.74$ (f) $d = 1.94$

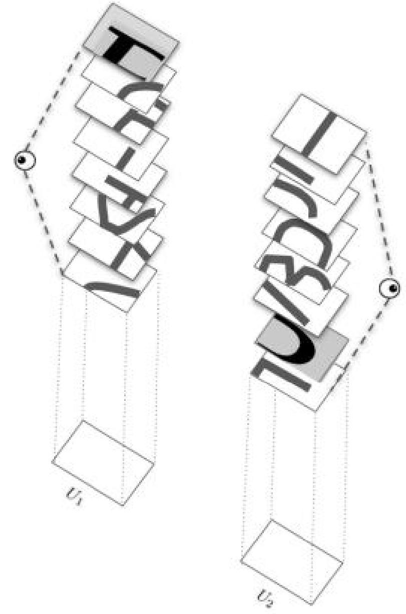
(g) $d = 2.00$

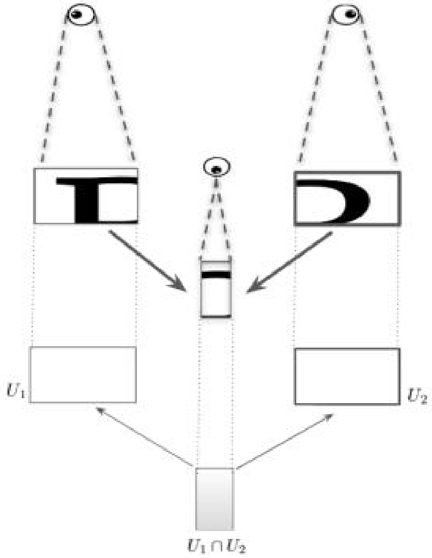


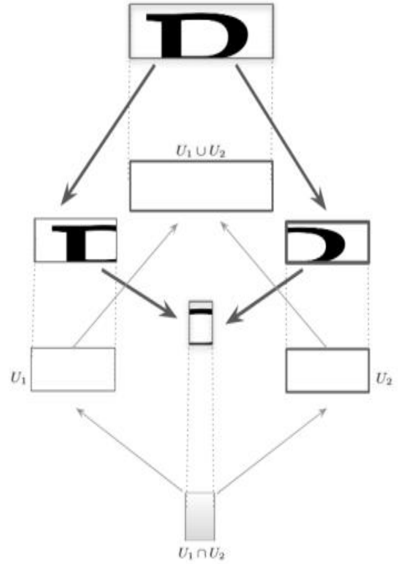
Sheaf

Definition 3.1.3. (*Definition of a Sheaf (again)*) Given a presheaf $F : \mathscr{O}(X)^{op} \to$ **Set**, an open set U with open cover by $\{U_i\}_{i \in I}$, and an I-indexed family $s_i \in F(U_i)$, then F is a sheaf provided it satisfies both:

1. (*Existence/Gluing*) If, for each i, there is a section $s_i \in F(U_i)$ satisfying that for each pair U_i and U_j the restrictions of s_i and s_j to the overlap $U_i \cap U_j$ match (or are "compatible")—in the sense that
$$s_i x = s_j x$$
for all $x \in U_i \cap U_j$ and all i, j—then there *exists* a section $s \in F(U)$ with restrictions $s|_{U_i} = s_i$ for all i. (Here, such an s is then called the *gluing*, and the s_i are called *compatible*.)
2. (*Uniqueness/Locality*) if $s, t \in F(U)$ are such that
$$s|_{U_i} = t|_{U_i}$$
for all i, then $s = t$. (In other words, there is *at most one* s with restrictions $s|_{U_i} = s_i$.)







Cellular Sheaf

- Mathematically, a cellular sheaf of $\mathbb{R}$-vector spaces on an abstract simplicial complex K is defined as a functor $\mathcal{F}\colon (K, \leq) \to Vect_{\mathbb{R}}$, from the poset category $(K, \leq)$ to the category $Vect_{\mathbb{R}}$ of vector spaces and linear transformations, where $\leq$ denotes the partial order induced by the face relations of simplices in K.
- Specifically, a cellular sheaf $\mathcal{F}\colon (K, \leq) \to Vect_{\mathbb{R}}$ over an abstract simplicial complex K consists of the following data:

a. A vector space $\mathcal{F}_\sigma$ for each simplex $\sigma \in K$ (these vector spaces are called stalks);

b. An $\mathbb{R}$-linear transformation $\mathcal{F}_{\sigma,\tau}\colon \mathcal{F}_\sigma \to \mathcal{F}_\tau$ for simplices $\sigma \leq \tau$, satisfying $\mathcal{F}_{\sigma,\sigma}\colon \mathcal{F}_\sigma \to \mathcal{F}_\sigma$ is defined as the identity map, and $\mathcal{F}_{\tau,\eta} \circ \mathcal{F}_{\sigma,\tau} = \mathcal{F}_{\sigma,\eta}$ for every triple towered simplices $\sigma \leq \tau \leq \eta$ in K.

- For a graph $G = (V, E)$, a cellular sheaf $\mathcal{F}\colon (G, \leq) \to Vect_{\mathbb{R}}$ can be defined as follows:

a. A vector space $\mathcal{F}_v$ for each vertex $v \in V$;

b. A vector space $\mathcal{F}_e$ for each edge $e \in E$;

c. A linear transformation $\mathcal{F}_{v,e}\colon \mathcal{F}_v \to \mathcal{F}_e$ for any ordered pair $v \leq e$.

Sheaf Laplacion

- To define the sheaf cohomology and sheaf Laplacian of $\mathcal{F}$, the cochain spaces $C^q(K,\mathcal{F}) = \bigoplus_{\sigma \epsilon K_{(q)}} \mathcal{F}_\sigma$ and the coboundary maps $\delta^q : C^q(K,\mathcal{F}) \rightarrow C^{q+1}(K,\mathcal{F})$ are essential components.
- The signed incidence function $[\cdot : \cdot] : K \times K \rightarrow \{-1, 0, 1\}$

$$[\sigma : \tau] = \begin{cases} (-1)^i & , \text{ if } \tau = [v_0, v_1, ..., v_n] \text{ and } \sigma = [v_0, ..., \widehat{v_i}, ..., v_n], \\ 0 & , \text{ otherwise.} \end{cases}$$

- The q-th coboundary map $\delta^q : C^q(K,\mathcal{F}) \rightarrow C^{q+1}(K,\mathcal{F})$ is thus defined as the $\mathbb{R}$-linear map extended by the following assignment:

$$\delta^q|_{\mathcal{F}_\sigma} = \sum_{\tau \in K_{(q+1)}} [\sigma : \tau] \cdot \mathcal{F}_{\sigma,\tau}$$

- The q-th sheaf cohomology is defined as the quotient vector space $H^q(K,\mathcal{F}) = \dfrac{\ker(\delta^q)}{\operatorname{im}(\delta^{q-1})}$.
- By considering each δ^q as a matrix B_p over $\mathbb{R}$, the q-th sheaf Laplacian of the cochain complex $C^*(K,\mathcal{F})$ is

$$\Delta_q = B_q^T B_q + B_{q-1} B_{q-1}^T$$

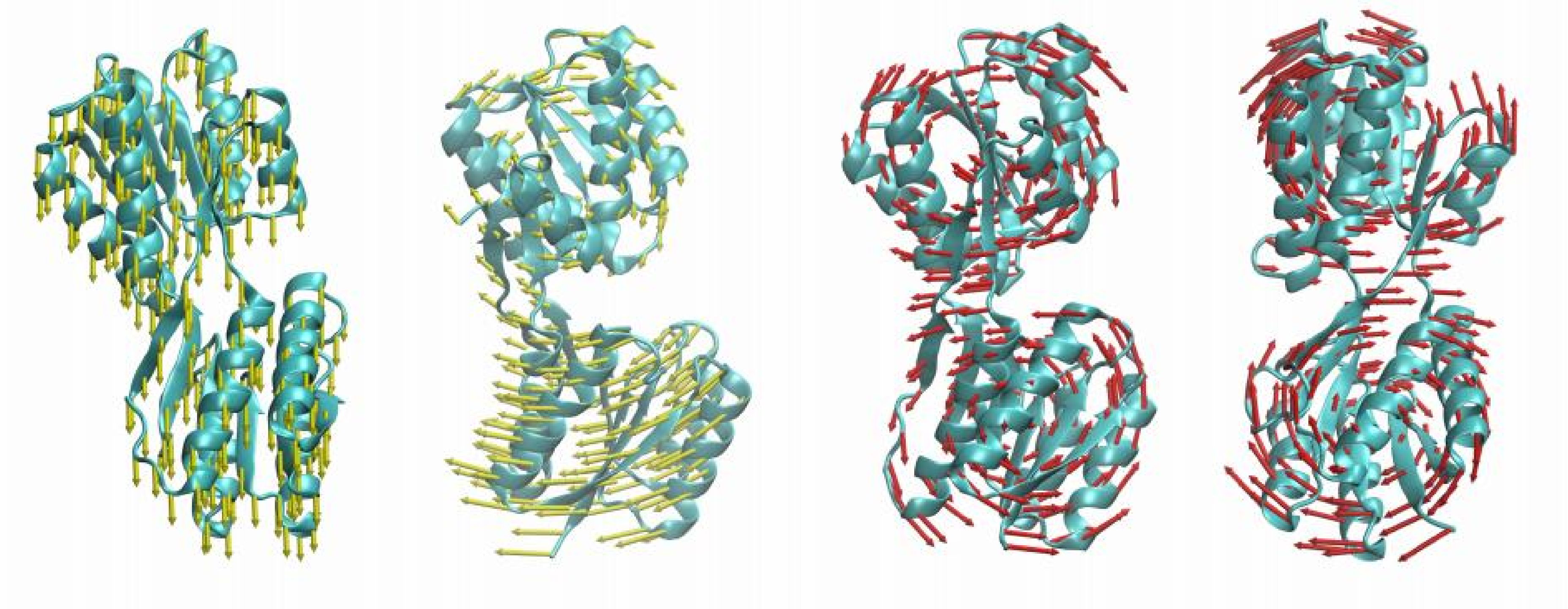
求是創新

Sheaf Laplacion

- By considering each δ^q as a matrix B_p over $\mathbb{R}$, the q-th sheaf Laplacian of the cochain complex $C^*(K,\mathcal{F})$ is

$$\Delta_q = B_q^T B_q + B_{q-1} B_{q-1}^T$$

- Celluar sheaf of a graph can be regarded as graph structure with extra information. The sheaf Laplacion spectrum analysis can provide graph features for data analysis and deep learning.



Eigenmodes with zero eigenvalues　　Eigenmodes with non-zero eigenvalues

Sheaf Neural Networks (NeurIPS 2020)

Sheaf Neural Networks

Jakob Hansen*
Department of Mathematics
The Ohio State University
hansen.612@osu.edu

Thomas Gebhart*
Department of Computer Science
University of Minnesota
gebhart@umn.edu

Abstract

We present a generalization of graph convolutional networks by generalizing the diffusion operation underlying this class of graph neural networks. These *sheaf neural networks* are based on the *sheaf Laplacian*, a generalization of the graph Laplacian that encodes additional relational structure parameterized by the underlying graph. The sheaf Laplacian and associated matrices provide an extended version of the diffusion operation in graph convolutional networks, providing a proper generalization for domains where relations between nodes are non-constant, asymmetric, and varying in dimension. We show that the resulting sheaf neural networks can outperform graph convolutional networks in domains where relations between nodes are asymmetric and signed.

1 Introduction

Graph neural networks are a class of generalized neural network architectures that take as input relational data and learn to classify the input graph or its nodes. Because this relational data lacks a constant local Euclidean structure, the definition of a local convolution-type operator through which weight sharing may be achieved is a non-trivial architectural challenge. A natural approach, inspired by the theory of graph signal processing [8] is to define convolution via the graph Laplacian or adjacency matrices [1, 3, 7] so that the layer-wise convolution operation acts as neighborhood

04

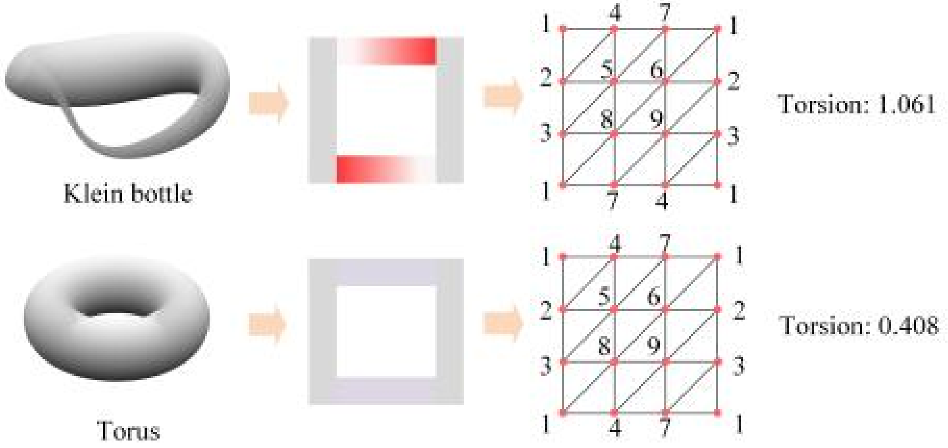
Torsion and Tor Functor:

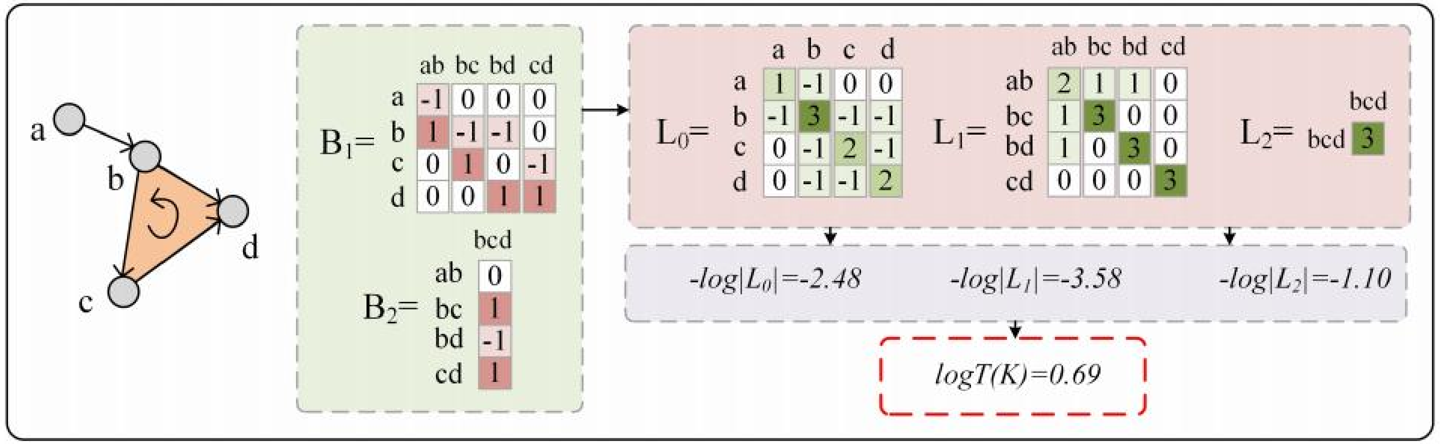
can abstract algebra helps data analysis?

Torsion

- The Reidemeister torsion, or R-torsion, was proposed by Kurt Reidemeister in 1935. It is the first topological invariant that could distinguish spaces with the same homotopy type.
- For a simplicial complex K, we can compute its analytic torsion $T(K)$ using its Laplacian matrix $L_p = B_p^T B_p + B_{p+1} B_{p+1}^T$ where B_p is the p-th boundary matrix.

$$\log T(K) = \frac{1}{2} \sum_{p=0}^{n} (-1)^{p+1} p \log |L_p|$$

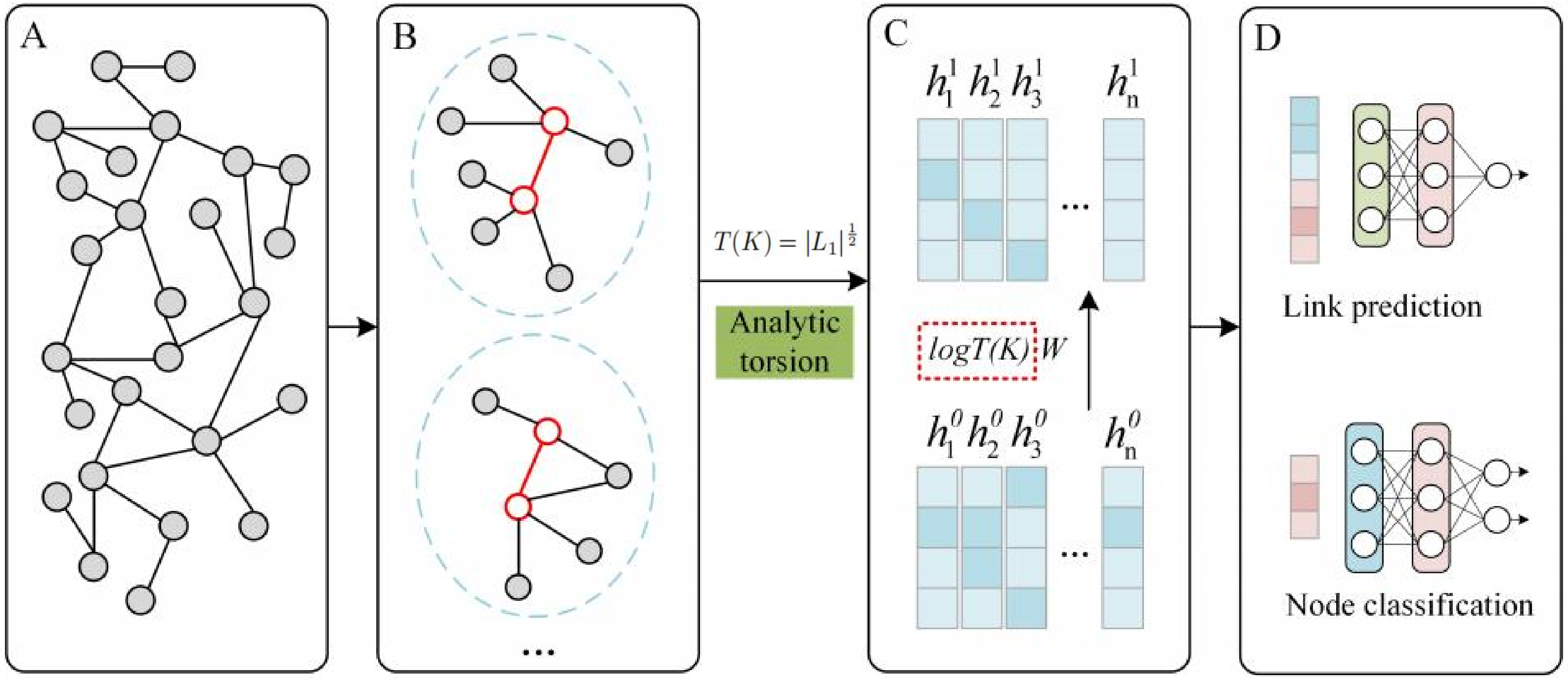




Torsion Graph Neural Network (TPAMI)

- Oliver Knill defined the analytic torsion for graphs (a special type of simplicial complex) through spectral properties of graph Laplacians, offering a new topological invariant for network analysis.
- Analytic torsion provides a powerful representation of the topological and geometric information within the data.

$$h_x^l = \sigma\left(\sum_{y\in\mathcal{N}(x)\cup\{x\}} \frac{1}{\sqrt{d(x)}\sqrt{d(y)}} \left|\log T(K_{x,y})\right| W_{\mathrm{GNN}} h_y^{l-1}\right)$$



Tor Functor and Tor-algebra

- Given an A-module M (A is a commutative finitely generated $\mathbb{F}$-algebra with unit, graded by non-negative even numbers and connected), a free resolution of M is an exact sequence of free A-modules

$$\cdots \xrightarrow{d_{i+1}} R^{-i} \xrightarrow{d_i} \cdots \xrightarrow{d_2} R^{-1} \xrightarrow{d_1} R^0 \xrightarrow{d_0} M \to 0.$$

- Assume we have a free resolution of an A-module M, and N is another A-module. Applying the functor $\otimes_A N$ to the free resolution, we obtain a cochain complex

$$\cdots \to R^{-i} \bigotimes_A N \to \cdots \to R^{-1} \bigotimes_A N \to R^0 \bigotimes_A N \to 0.$$

- The i-th cohomology module of the cochain complex is denoted as $Tor_A^i(M,N)$.

$$\mathrm{Tor}_A(M,N) = \bigoplus_{i,j>=0} \mathrm{Tor}_A^{-i,2j}(M,N)$$

Tor-algebra of simplicial complex

- Given a simplicial complex K, its face ring is denoted as $\mathbb{F}[K]$. We have

$$\mathrm{Tor}^{-i,2j}_{\mathbb{F}[v_1,v_2,\ldots,v_m]}(\mathbb{F}[\mathcal{K}],\mathcal{K}) = \bigoplus_{J\subset[m],|J|=j} \hat{H}^{j-i-1}(\mathcal{K}_J,\mathbb{F})$$

  where K_J is any full subcomplex of K obtained by restricting to $J \subset [m]$ and $\widehat{H}(K_J, \mathbb{F})$ is the reduced simplicial cohomology of K_J.

- For a filtration process, we can get a sequence of Tor-algebra

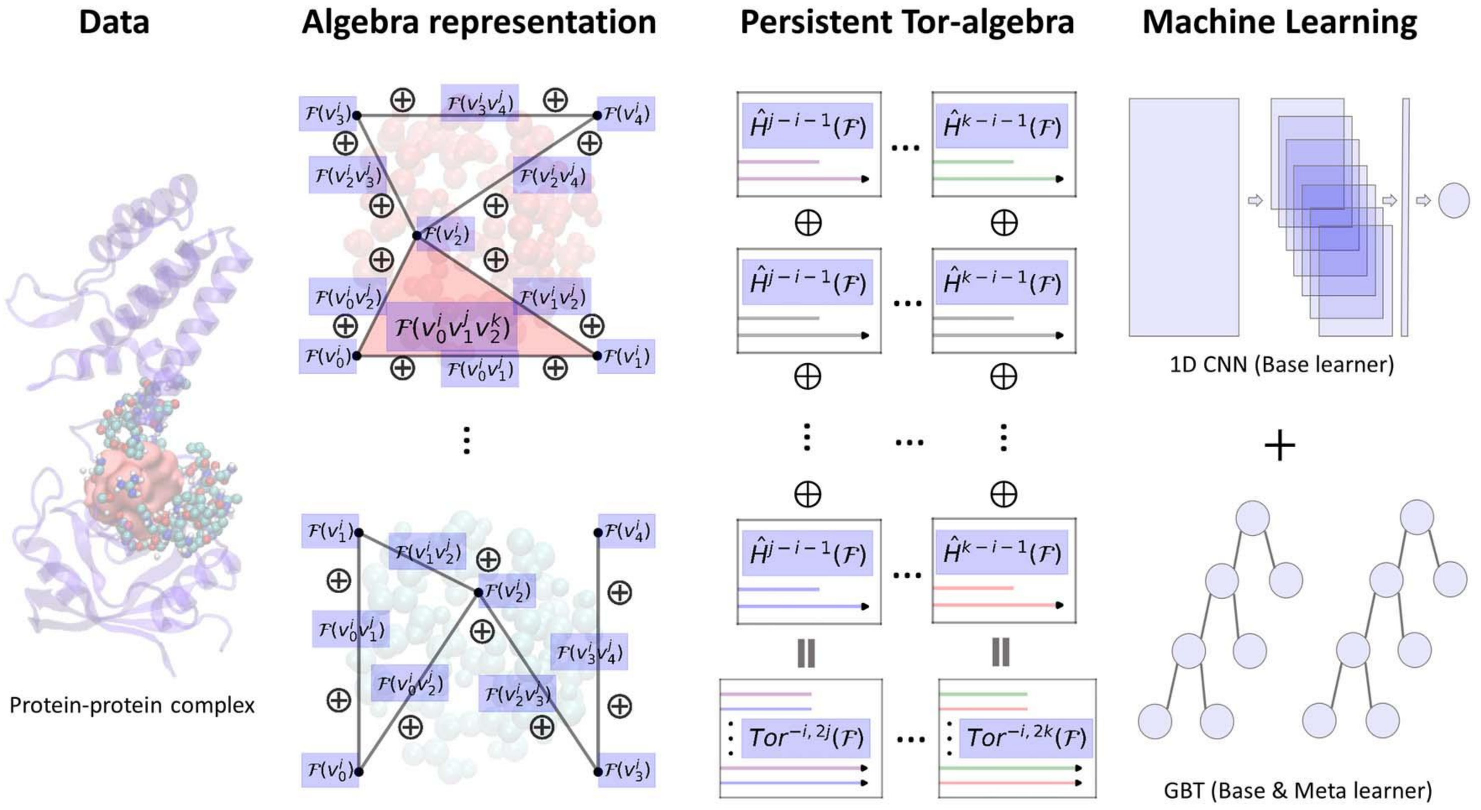
$$\mathrm{Tor}(\mathcal{K}_0) \leftarrow \mathrm{Tor}(\mathcal{K}_1) \leftarrow \ldots \leftarrow \mathrm{Tor}(\mathcal{K}_n),$$

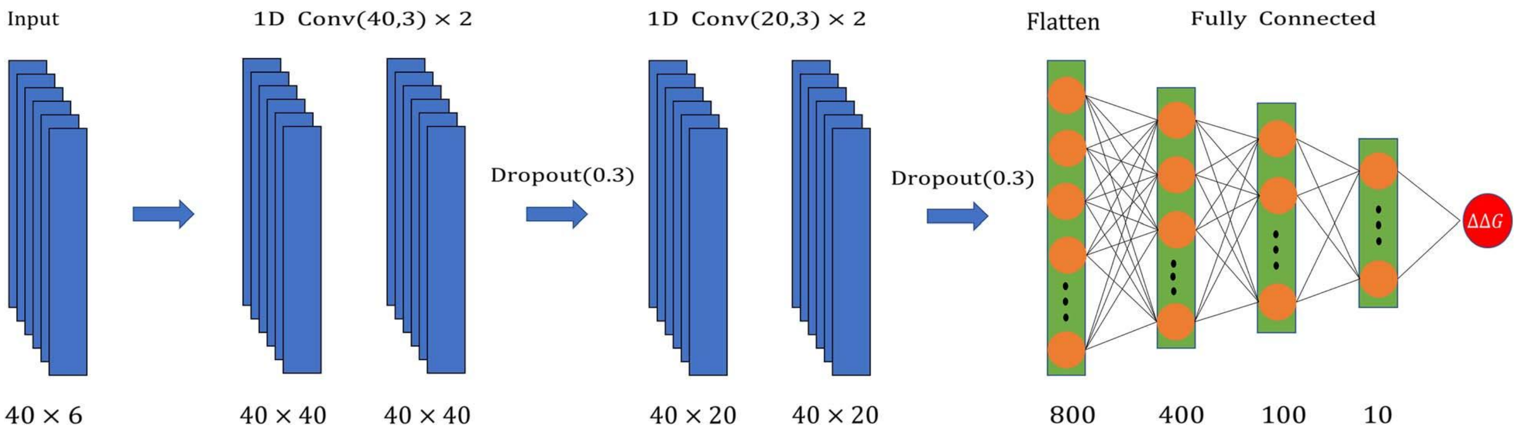
- Further, by considering a specific graded component of Tor-algebra modules, we can get the persistent graded Tor-algebra for a given $(-i, 2j)$. We have

$$\mathrm{Tor}^{-i,2j}(\mathcal{K}_1) \leftarrow \mathrm{Tor}^{-i,2j}(\mathcal{K}_2) \leftarrow \ldots \leftarrow \mathrm{Tor}^{-i,2j}(\mathcal{K}_n).$$

  which can be computed using reduced cohomology.

Tor-based Deep Learning





求是創新

Application of Commutative Algebra

COMMUTATIVE ALGEBRA-ENHANCED TOPOLOGICAL DATA ANALYSIS

CHUAN-SHEN HU, YU WANG, KELIN XIA, KE YE, AND YIPENG ZHANG

ABSTRACT. Topological Data Analysis (TDA) combines computational topology and data science to extract and analyze intrinsic topological and geometric structures in data set in a metric space. While the persistent homology (PH), a widely used tool in TDA, which tracks the lifespan information of topological features through a filtration process, has shown its effectiveness in applications, it is inherently limited in homotopy invariants and overlooks finer geometric and combinatorial details. To bridge this gap, we introduce two novel commutative algebra-based frameworks which extend beyond homology by incorporating tools from computational commutative algebra : (1) *the persistent ideals* derived from the decomposition of algebraic objects associated to simplicial complexes, like those in theory of edge ideals and Stanley–Reisner ideals, which will provide new commutative algebra-based barcodes and offer a richer characterization of topological and geometric structures in filtrations. (2)*persistent chain complex of free modules* associated with traditional persistent simplicial complex by labelling each chain in the chain complex of the persistent simplicial complex with elements in a commutative ring, which will enable us to detect local information of the topology via some pure algebraic operations. *Crucially, both of the two newly-established framework can recover topological information got from conventional PH and will give us more information.* Therefore, they provide new insights in computational topology, computational algebra and data science.

1. INTRODUCTION

Topological Data Analysis (TDA) is an interdisciplinary field that integrates computational topology, homological algebra, and data science to uncover and analyze intrinsic topology and geometry within high-dimensional, noisy, and complex datasets [6, 23, 8, 54, 18]. By leveraging algebraic topology, TDA provides a robust framework for detecting and quantifying topological structures. Due to its ability to extract meaningful topological information from structured and unstructured data, TDA has found applications across various scientific and engineering areas such as bioinformatics [32, 39, 3], material science [47, 30, 46], and deep and machine learning models [56, 39, 9].

05

Summary and Discussion

Summary and Discussion

➢ By discretization, basic tools in fundamental mathematics can be utilized to data analysis and deep learning. These tools can extract some important intrinsic features from data.

➢ However, what are the practical meanings of these features derived from fundamental mathematics?

浙江大學
ZHEJIANG UNIVERSITY

Thank you!